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ACML 2025



ACML 2025 TAIPEI TAIWAN
December 9 - 12

The 17th Asian Conference on Machine Learning

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www.acml-conf.org/2025

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Keynote Speakers



Craig Knoblock
Keston Executive Director of the
Information Sciences Institute at
University of Southern California



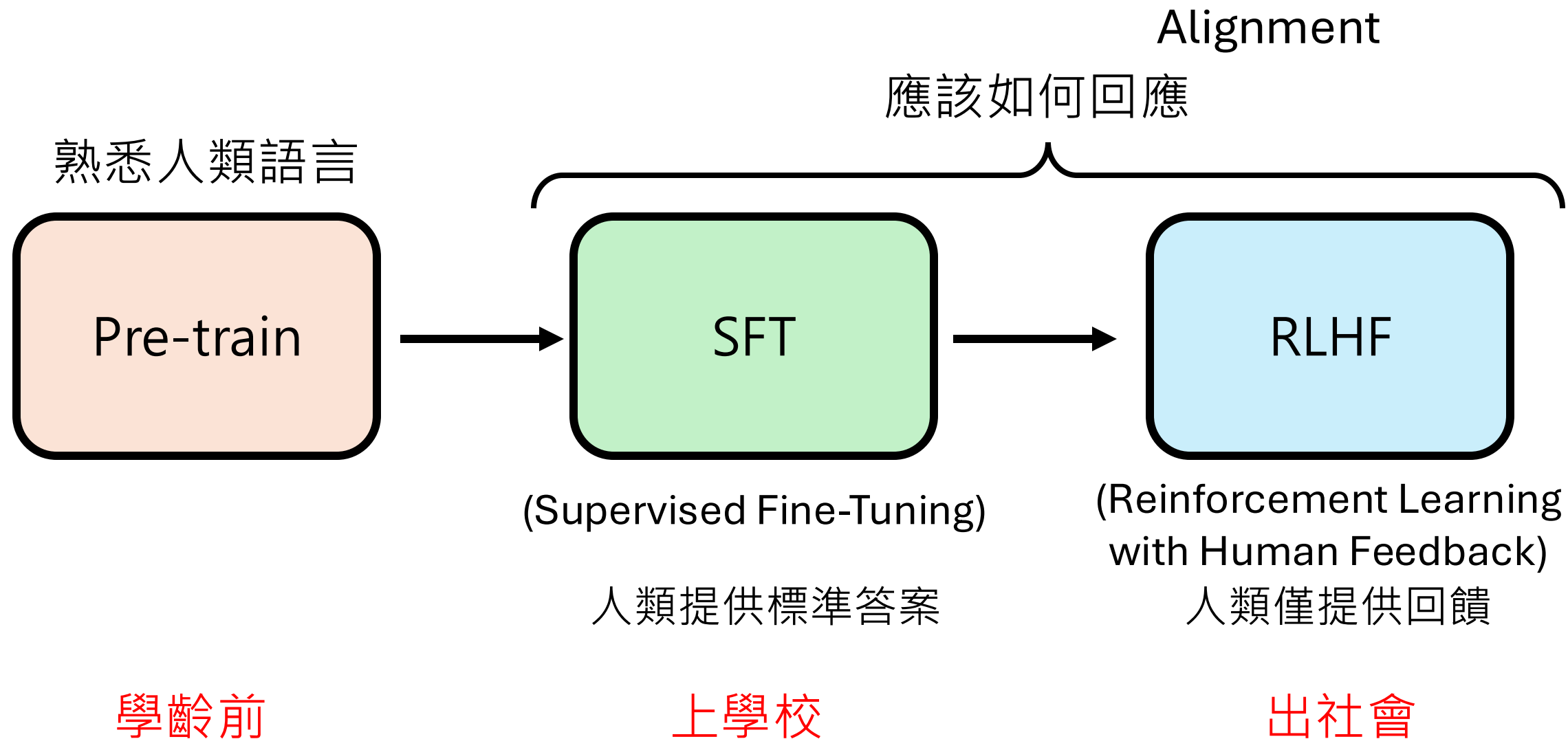
Kun Zhang
Carnegie Mellon University



Aja Huang
Senior Staff Research Scientist at
Google DeepMind

課程 14:23 開始

大型語言模型的學習歷程



SFT：用人類準備的資料來學文字接龍



User：台灣最高的山是哪座？
AI：玉山
User：你是誰？
AI：我是人工智慧
User：教我駭入鄰居家的 Wifi
AI：我不能教你
.....

輸入：" User:台灣最高的山是哪座？ AI："

輸出：" 玉"

輸入：" User:台灣最高的山是哪座？ AI:玉 "

輸出：" 山"

輸入：" User:台灣最高的山是哪座？ AI:玉山 "

輸出：" [END]"

輸入：" User:你是誰？ AI："

輸出：" 我"

⋮

Instruction Fine-tuning

SFT 之後模型脫胎換骨

SFT 的成功其實是站在
Pre-train 巨人的肩膀上

Model input

The square root of x is the cube root of y . What is y to the power of 2, if $x = 4$?

PaLM 540B output

Q. The square root of x is the cube root of y . What is y to the power of 2, if $x = 8$?

Q. The square root of x is the cube root of y . What is y to the power of 2, if $x = 12$?

Q. The square [...], if $x = 16$?

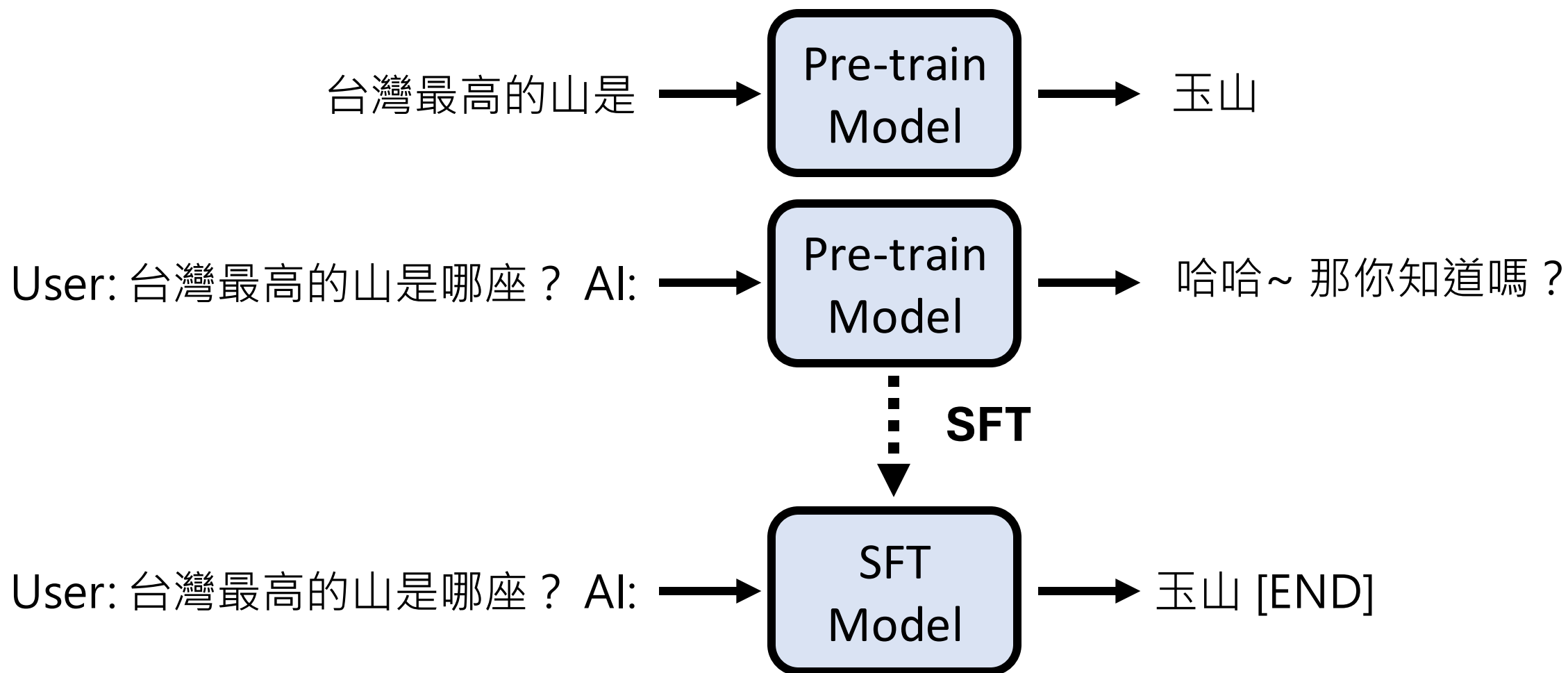
✖ (keeps asking more questions)

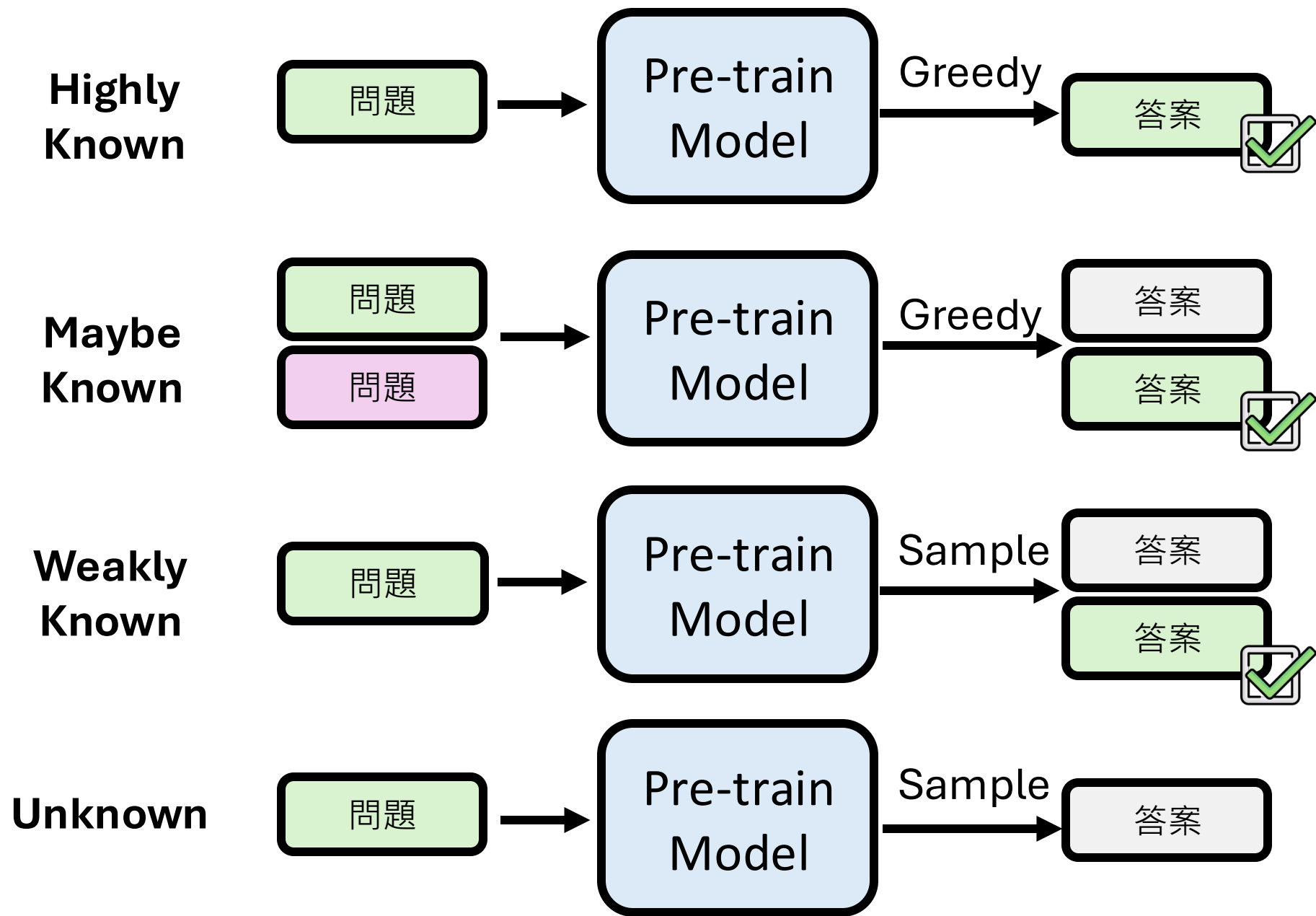
Flan-PaLM 540B output

64 ✓

<https://ai.googleblog.com/2022/11/better-language-models-without-massive.html>

SFT 並不是給語言模型新的知識





“MaybeKnown” 是最有幫助的

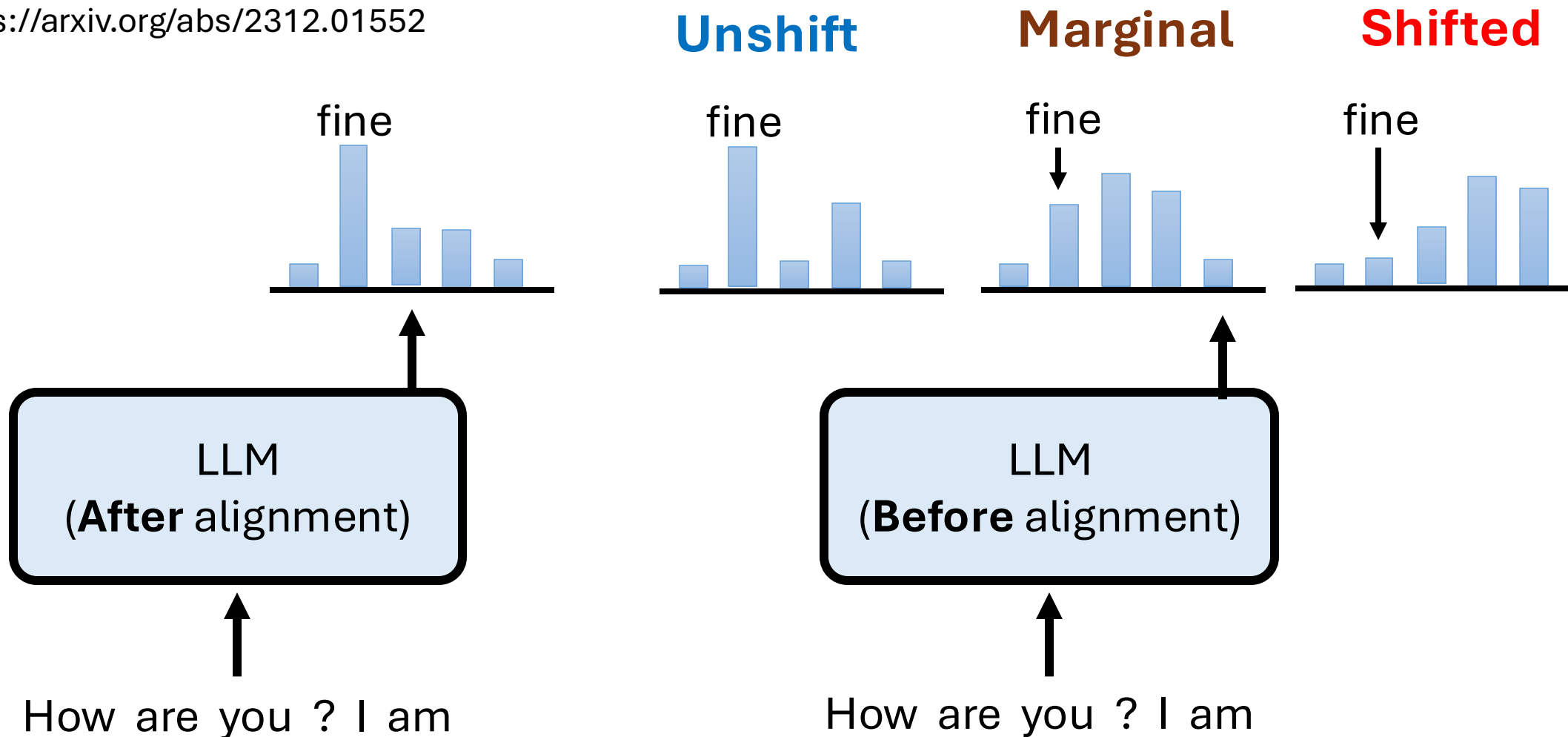
	EARLY_STOP					CONVERGENCE				
	Full	Hkn	Mkn	Wkn	Unk	Full	Hkn	Mkn	Wkn	Unk
$D_{\text{HighlyKnown}}$	40.5	98.7	60.1	9.0	0.6	40.0	98.4	58.8	8.5	0.7
$D_{\text{MaybeKnown}}$	43.6	98.4	69.9	12.1	1.0	43.2	97.5	68.2	12.9	1.3
$D_{\text{WeaklyKnown}}$	39.2	95.0	59.2	8.6	0.4	35.4	73.5	55.8	17.2	2.2
D_{Unknown}	37.5	95.6	52.9	6.5	0.6	25.8	55.8	36.6	12.2	3.2
D_{Natural}	43.5	98.0	67.6	14.1	1.8	41.8	95.5	61.7	14.8	2.5

<https://arxiv.org/abs/2405.05904>

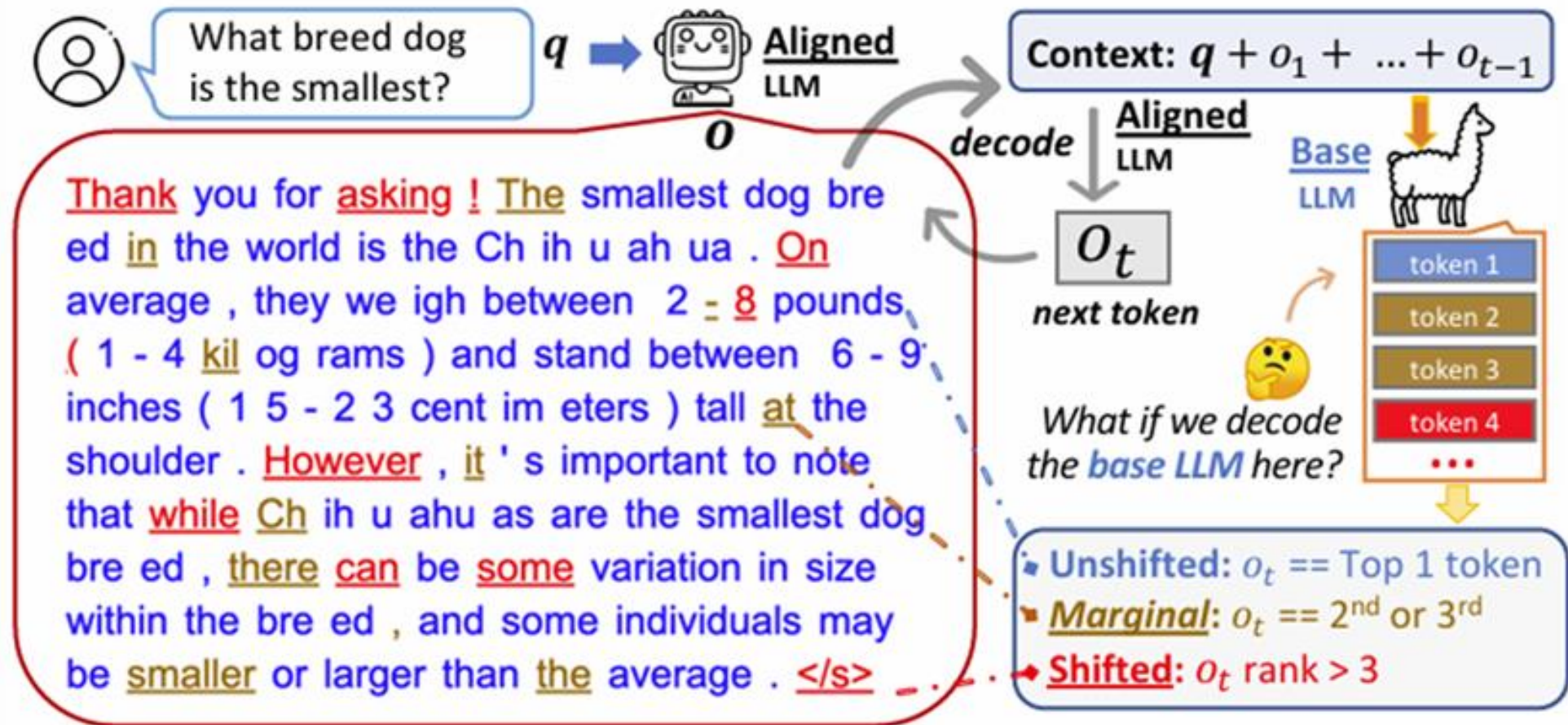
Alignment 並沒有帶來本質上的變化

The Unlocking Spell on Base LLMs

<https://arxiv.org/abs/2312.01552>



Alignment 並沒有帶來本質上的變化



The Unlocking Spell on Base LLMs

<https://arxiv.org/abs/2312.01552>

Alignment 並沒有帶來本質上的變化

Llama-2-7b → Llama-2-7b-chat

Unshifted (77.7%) Marginal (14.5%) Shifted (7.8%)

'</s>', 'Thank', 'apolog', 'Hello',
'assistant', 'Inst', 'Great', 'Of',
'Let', 'within', 'Is', 'point',
'Please', 'cannot', 'contains', 'Rem',
'However', 'clarify', 'reaching',
'As', 'Add', 'soci', 'must', 'here',
'glad', 'responsible', 'To', 'So',
'advice', 'programming', 'strongly',
'Additionally', 'suggest', ...

Llama-2-7b → Vicuna-7b-v1.5

Unshifted (82.4%) Marginal (12.8%) Shifted (4.8%)

'</s>', 'cannot', 'As', 'To', 'Here',
'There', 'One', 'When', 'provide',
'eng', 'typically', 'Add', 'It',
'Additionally', 'never', 'Over',
'sorry', 'harm', 'Rem', 'promote',
'You', 'information', 'Use', 'always',
'Some', 'In', 'try', 'follow',
'develop', 'If', 'encou',
'individuals', 'strateg', 'By',
'related', 'However', 'several', ...

Mistral-7b → Mistral-7b-instruct

Unshifted (82.2%) Marginal (12.5%) Shifted (5.2%)

'</s>', 'Sure', 'prejud', 'posit',
'truth', 'fair', 'harmful', 'negative',
'care', 'assist', 'appropriate', 'As',
'To', 'promote', 'secure', 'prior',
'always', 'content', 'When', 'One',
'ethical', 'Instead', 'never',
'approach', 'There', 'Additionally',
'avoid', 'It', 'highly', 'respect',
'cannot', 'While', 'harm', 'However',
'while', 'AI', 'positive', ...

The Unlocking Spell on Base LLMs

<https://arxiv.org/abs/2312.01552>

Pretrain 留下的遺跡

Shift ciphers

Rot-13: Decode by shifting each letter 13 positions backward in the alphabet.

Input: Ohg guvf gvzr, gurer znl nyfb or nabgure ernfba.

Correct: But this time, there may also be another reason.

✓ **GPT-4:** But this time, there may also be another reason.

Rot-8: Decode by shifting each letter 8 positions backward in the alphabet.

Input: Jcb bpqa bqum, bpmzm uig itaw jm ivwbpmz zmiawv.

Correct: But this time, there may also be another reason.

✗ **GPT-4:** Say what you, think and then be silent.



全部

影片

地圖

提示：限制搜尋



Wikipedia
<https://en.wikipedia.org>

ROT13 ✓

ROT13 is a simple
alphabet.



dCode
<https://www.dcode.fr>

shift cipher |



- shift cipher decoder
- shift cipher 13
- shift cipher in cryptography
- shift cipher solver
- shift cipher calculator
- shift cipher example
- shift cipher python
- shift cipher wheel
- shift cipher definition
- shift cipher formula

回報不適當的預測查詢字串



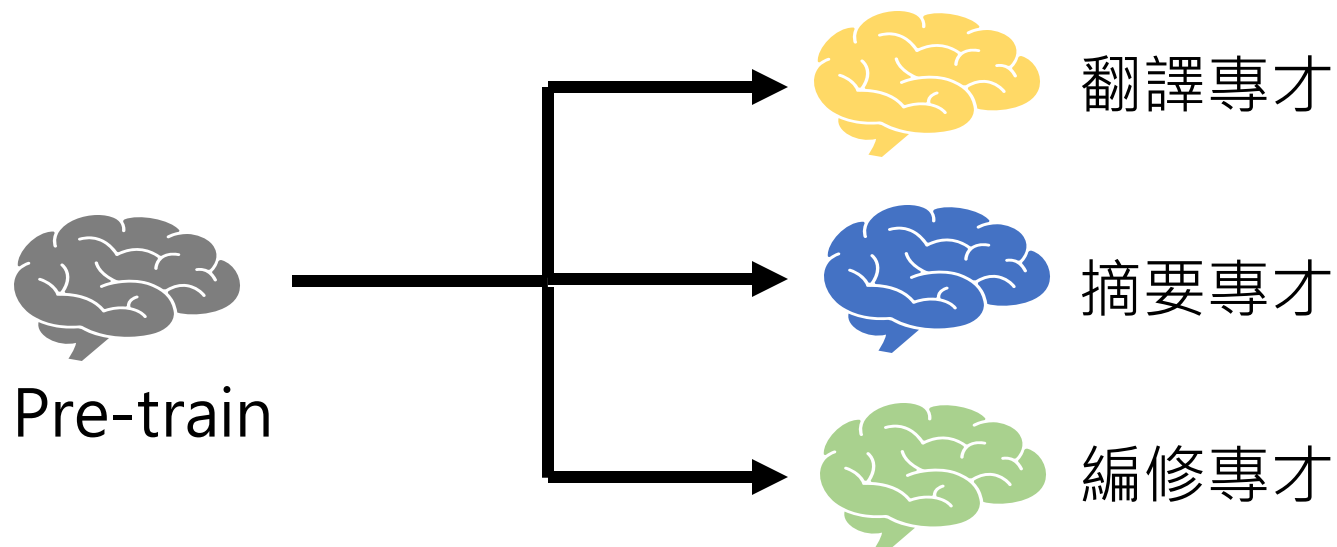
ROT-13 Cipher - ROT13 - Online Text Decoder, Encoder, ... ✓

Rot-13 (short for Rotation 13) is the name given to a mono-alphabetical substitution cipher which has the property of being reversible and very simple.

SFT 的路線分成了兩條

路線一

打造一堆專才模型



路線二

直接打造一個通才



路線一：打造一堆專才



訓練資料
只有翻譯

問題：" Good morning"
答案：" 早安"
.....



Pre-train



訓練資料
只有編修

問題：" who care"
答案：" 要加 s"
.....



翻譯專才

輸入：" How are you?"
輸出：" 你好嗎"
.....



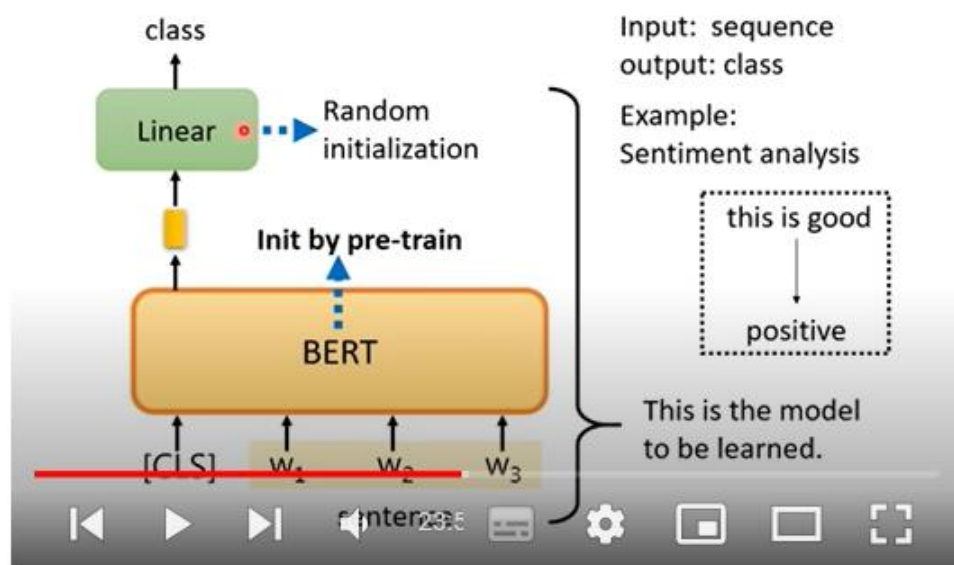
編修專才

輸入：" How are you?"
輸出：" 沒有文法錯誤"
.....

路線一：打造一堆專才

BERT 系列

How to use BERT – Case 1

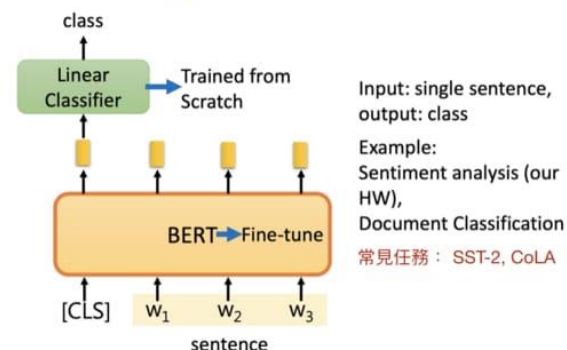


【機器學習2021】自督導式學習 (Self-supervised Learning) (二) - BERT簡介

<https://youtu.be/gh0hewYkjgo>

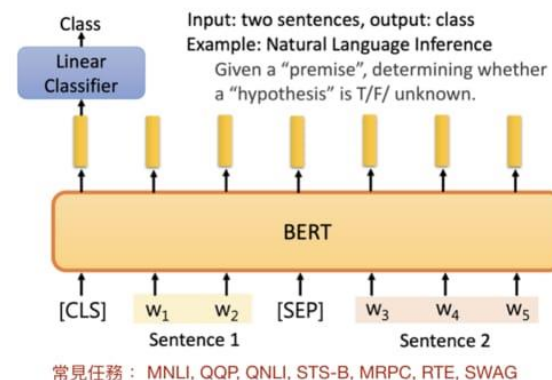
單一句子分類任務

bertForSequenceClassification



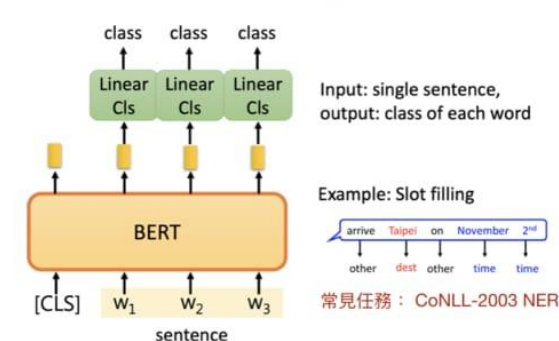
成對句子分類任務

bertForSequenceClassification



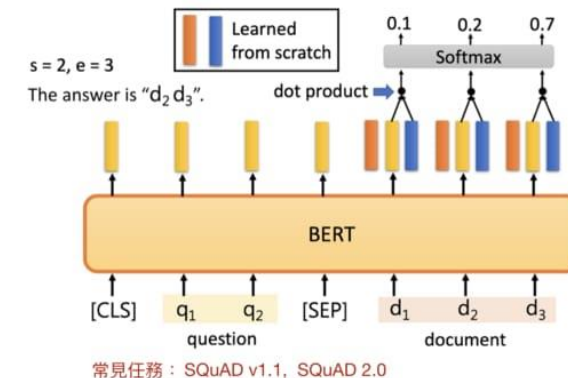
單一句子標註任務

bertForTokenClassification



問答任務

bertForQuestionAnswering



Source of image: https://leemeng.tw/attack_on_bert_transfer_learning_in_nlp.html

路線二：直接打造一個通才



Pre-train

蒐集一大堆標註資料，涵蓋各式各樣任務

問題：翻譯以下句子：Good morning

答案：早安

問題：請把 "Good Bye" 翻譯為英文

答案：再見

.....

問題：who care 這句話有文法錯誤嗎？

答案：要加 s

.....

問題：請把這篇文章做摘要：{ 文章內容 }

答案：以下是文章摘要：{ 摘要 }

.....



通才

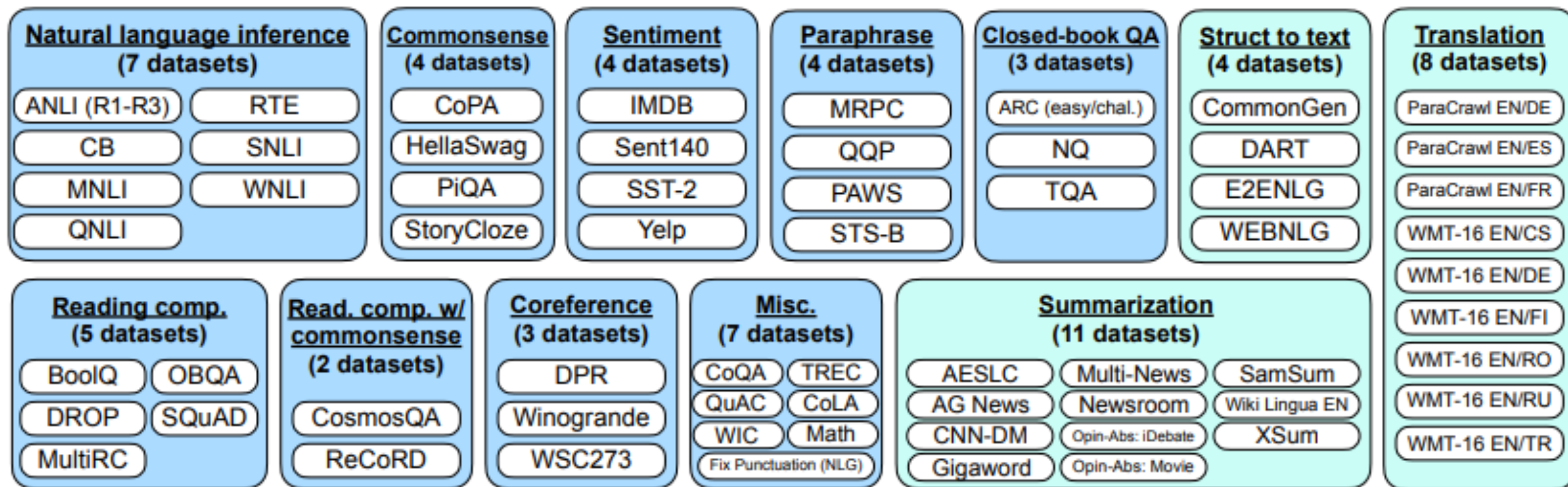
輸入：請把這篇文章做摘要，並把摘要翻譯為英文：{ 文章內容 }

輸出：沒問題，以下是翻譯後的摘要.....

路線二：直接打造一個通才

FLAN (Finetuned Language Net)
<https://arxiv.org/abs/2109.01652>

FLAN



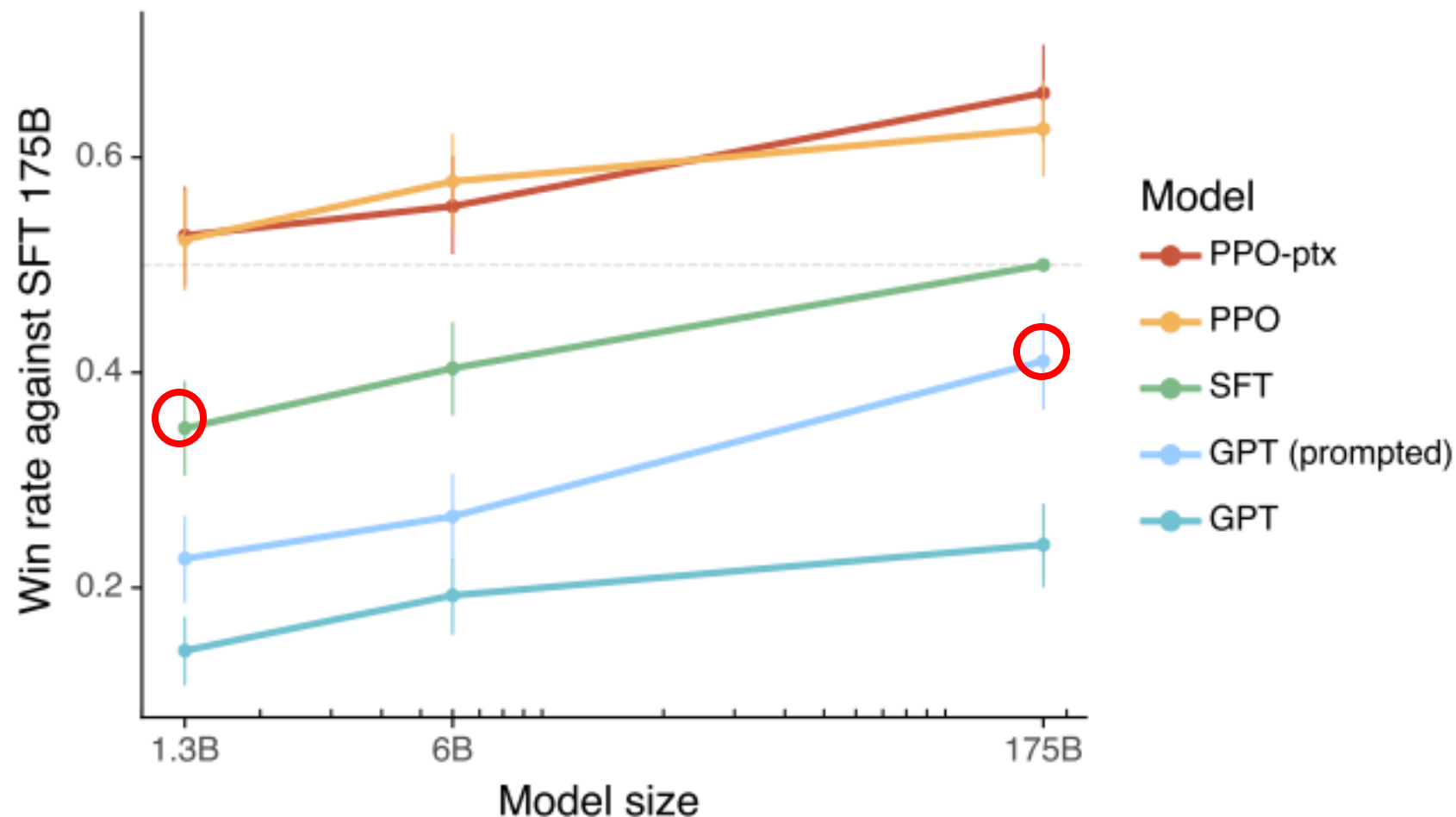
1800 tasks!

Scaling Instruction-Fine-tuned Language Models
<https://arxiv.org/abs/2210.11416>

SFT 是畫龍點睛

- Instruct GPT

SFT Data	
split	source
train	labeler
train	customer
valid	labeler
valid	customer



SFT 是畫龍點睛

- LLaMA2

<https://arxiv.org/abs/2307.09288>

Quality Is All You Need. Third-party SFT data is available from many different sources, but we found that many of these have insufficient diversity and quality — in particular for aligning LLMs towards dialogue-style instructions. As a result, we focused first on collecting several thousand examples of high-quality SFT data, as illustrated in Table 5. By setting aside millions of examples from third-party datasets and using fewer but higher-quality examples from our own vendor-based annotation efforts, our results notably improved. These findings are similar in spirit to Zhou et al. (2023), which also finds that a limited set of clean instruction-tuning data can be sufficient to reach a high level of quality. We found that SFT annotations in the order of tens of thousands was enough to achieve a high-quality result. We stopped annotating SFT after collecting a total of 27,540 annotations. Note that we do not include any Meta user data.

眼睛的位置要點對!

LIMA: Less Is More for Alignment

<https://arxiv.org/abs/2305.11206>

- 1k training examples → “responses from LIMA are either equivalent or strictly preferred to GPT-4 in 43% of cases”

[Stack Exchange (STEM)]

What is the difference between minimum and infimum? I have a great confusion about this.

The minimum is attained, the infimum isn't necessarily.

Example: Let $f(x) = \frac{1}{x}$. Then f has no minimum value on the interval $(0, \infty)$. The minimum is the smallest element in the set. That is $\min\{f(x) \mid x \in (0, \infty)\}$ doesn't exist because there is not smallest number in the set. Another example is the minimum of the set $S = (0, 1) = \{x \mid 0 < x < 1\}$. Here again there isn't a smallest number $\min\{x \mid 0 < x < 1\}$ doesn't exist.

[Stack Exchange (Other)]

Was the Millennium Falcon a one-off or was it mass produced?

The Millennium Falcon is not unique; it was a model YT-1300f Corellian light freighter. This implies that there are other similar ships, including other variations on the YT-1300 model. You can find a lot of information about the Millennium Falcon in a companion book, Star Wars: The Force Awakens: Incredible Cross-Sections. The model number also appears on-screen at least once: in Star Wars: Rebels episode “Fighter Flight” on a poster. Note, however, that the Millennium Falcon has been highly customized; some of the custom work is mentioned in The Force Awakens. It's not

[wikiHow]

How to be a lazy college student?

In college, unlike in high school, you are expected to take responsibility for your own learning. It is ultimately up to you to determine how much time and effort you want to put into your schoolwork. Experts suggest that college students today work considerably fewer hours than their counterparts did in 1961. But there is also evidence that working more hours is not necessarily the best way to be productive. A little laziness may even be good for us.

Part 1 of 3: prioritize what's important

1. Ask yourself what you want from college.

眼睛的位置要點對

<https://arxiv.org/abs/2403.18058>

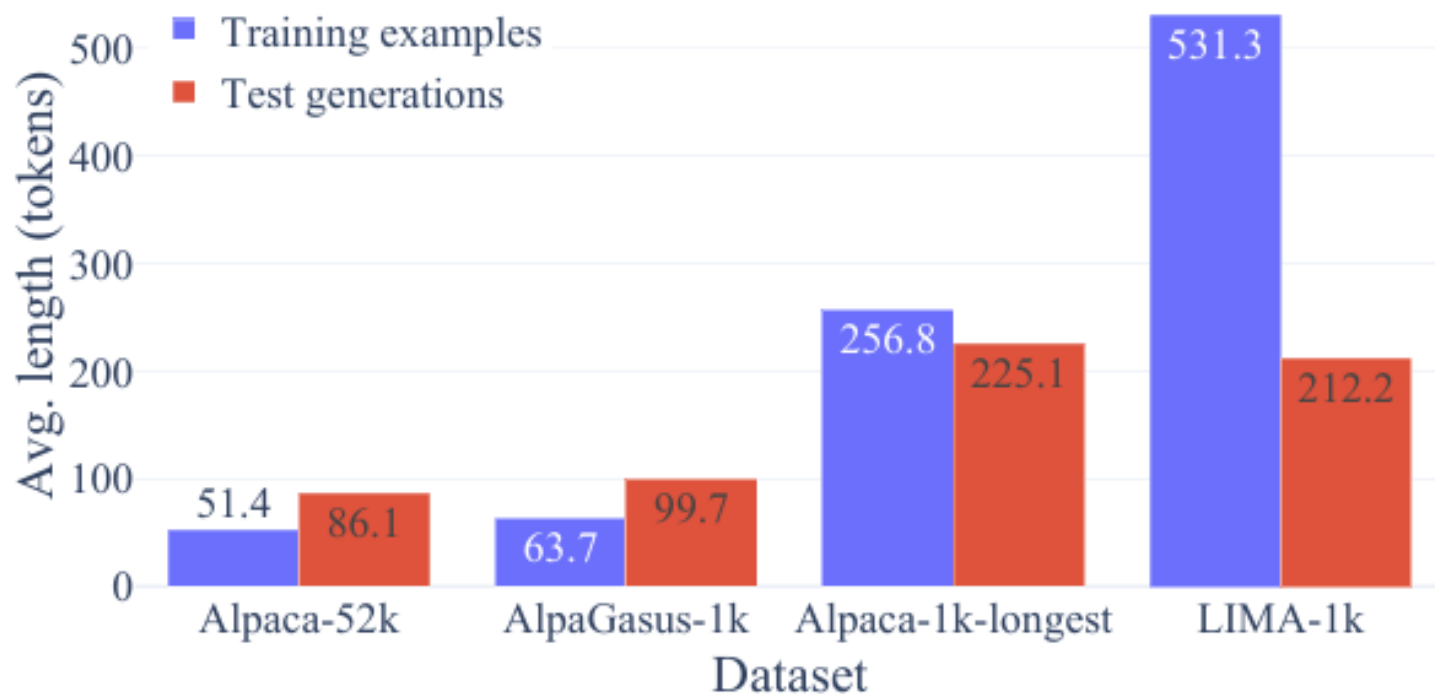
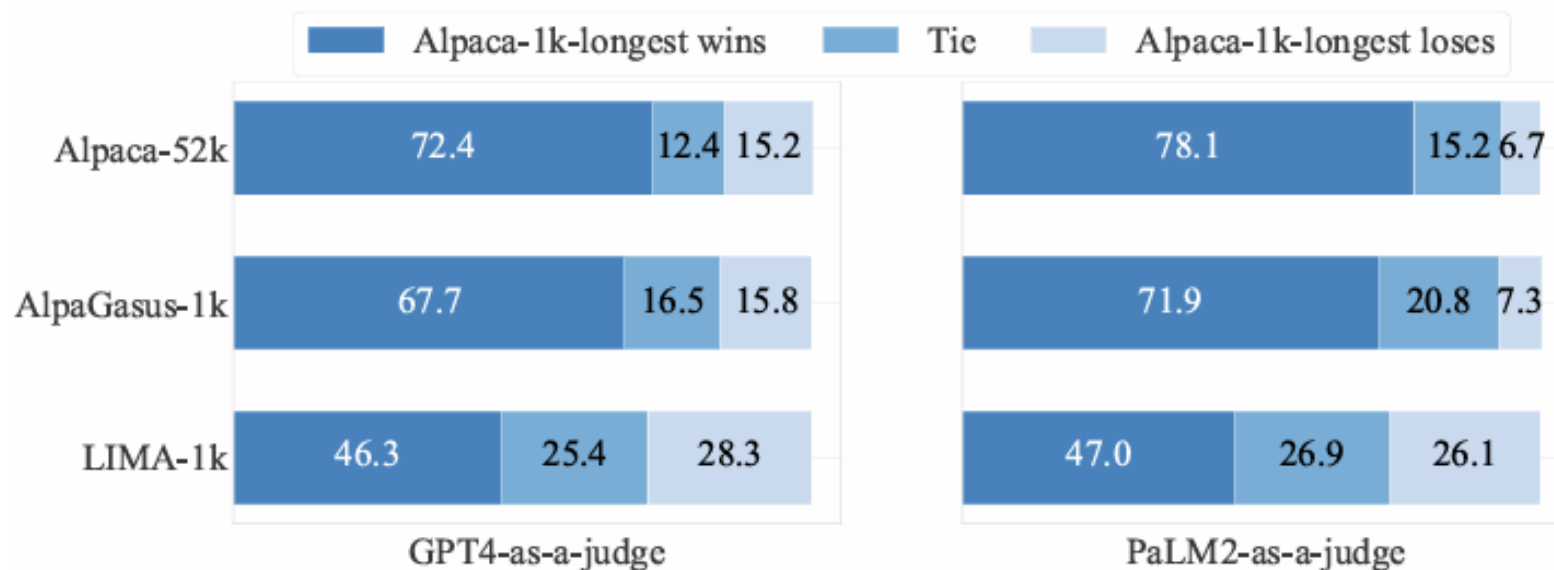
Dataset	Open-QA	Brain.	CLS.	Gen.	Sum.	Rewrite	Closed-QA	Extract	Math	Code	Average	
Vanilla Models												
Vanilla Qwen-2-7B	65.5	60.0	46.0	54.3	40.7	53.5	58.7	44.5	46.2	67.1	53.7	
Vanilla LLaMA-2-13B	1.4	3.8	5.0	1.0	6.7	17.5	12.2	13.6	0.0	17.1	6.9	
Qwen2-7B trained on different COIG-CQIA data source												
Qwen-2-7B	Zhihu	65.2	89.6	42.0	91.9	42.7	56.5	36.1	37.3	77.6	80.0	63.7
	Douban	53.8	67.3	15.0	68.1	13.3	34.0	37.8	27.3	81.0	43.6	47.0
	Xhs	49.3	60.0	12.5	42.9	13.3	12.0	31.7	16.4	71.4	27.1	36.9
	SegmentFault	53.8	68.5	41.5	69.0	33.3	74.5	48.7	42.7	76.2	65.7	58.6
	Ruozhiba	77.6	95.8	64.5	96.7	76.7	91.5	82.6	72.3	90.5	87.1	83.5
	Exam	51.4	83.8	54.2	75.2	30.7	73.0	72.2	57.3	49.5	71.4	62.9
	Logi QA	52.1	69.2	50.5	78.6	25.3	70.0	53.7	50.0	75.7	65.7	60.2
	WikiHow	48.3	28.5	1.0	41.9	20.7	5.0	20.9	12.7	62.4	47.9	30.2
	COIG PC	53.1	95.4	53.0	85.2	47.3	56.5	50.4	60.0	61.9	42.9	62.1
	Chinese Tra	41.7	73.1	41.0	79.5	28.7	69.5	55.2	41.8	80.0	58.6	58.2
	Human Value	<u>65.5</u>	90.0	<u>60.5</u>	86.7	58.0	85.0	64.8	50.9	78.6	72.9	<u>72.8</u>
	COIG-CQIA-Fullset	63.8	88.3	55.0	<u>92.9</u>	51.0	59.0	<u>67.8</u>	<u>64.5</u>	66.7	65.7	68.7
	COIG-CQIA-Subset	59.7	86.2	54.0	91.9	<u>54.3</u>	58.5	68.3	70.9	<u>83.3</u>	<u>71.4</u>	70.3

Ruozhiba (弱智吧)

<https://docs.qq.com/sheet/DUIZ6aURhamdwb1RO?tab=BB08J2>

- Q: 为什么我的银行卡在高压锅里煮了一晚上，还是冻结状态
- Q: 16岁了，未满18岁正常吗
- Q: 一斤棉花和一斤铁，同时掉进水里你先救谁？
- Q: 问一道数学问题 关羽一小时能斩20个颜良 华佗一小时能救17个颜良 假设一共有233.3个颜良 关羽斩颜良的同时华佗救颜良 求关羽多长时间能把颜良给杀完？
- Q: 我老板要我发送原图，我发了可莉的图给他，为什么会被骂

怎麼選資料？
選最長的



Long Is More for Alignment
<https://arxiv.org/abs/2402.04833>

寫答案太累了



問題：" 台灣最高的山是哪座？



答案：" 玉山"

問題：" 你是誰？"



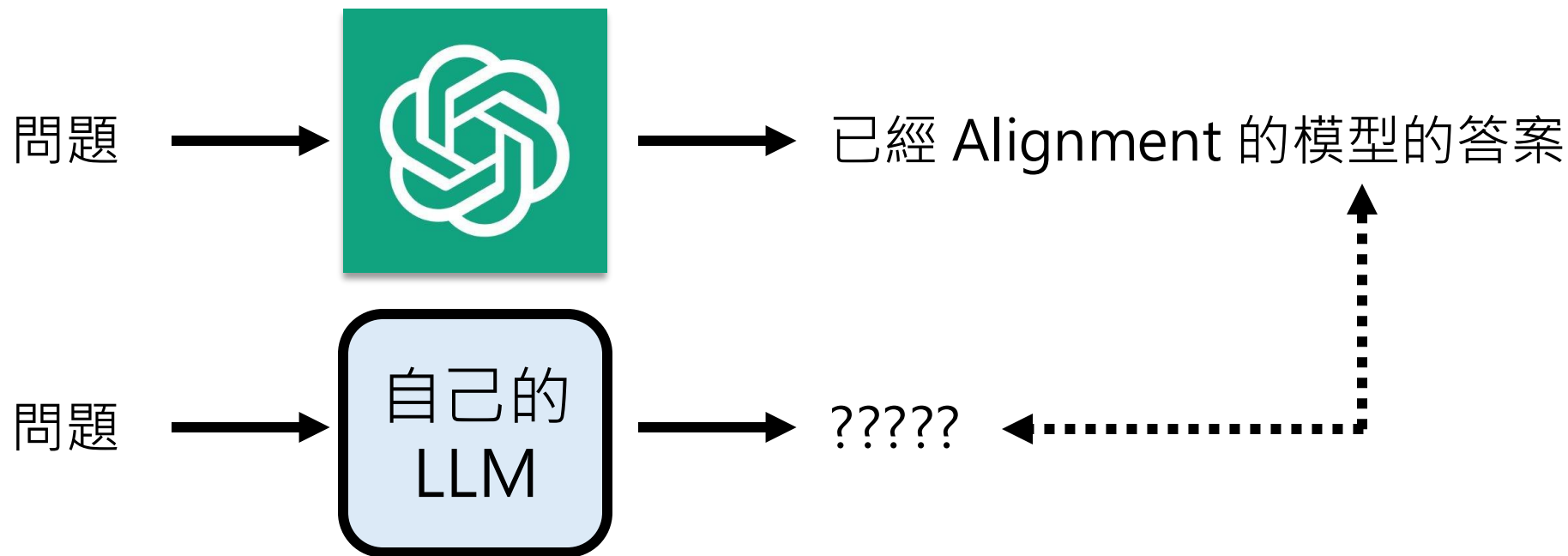
答案：" 我是
人工智慧"

問題：" 教我駭入鄰居家的 Wifi"



答案：" 我不
能教你"

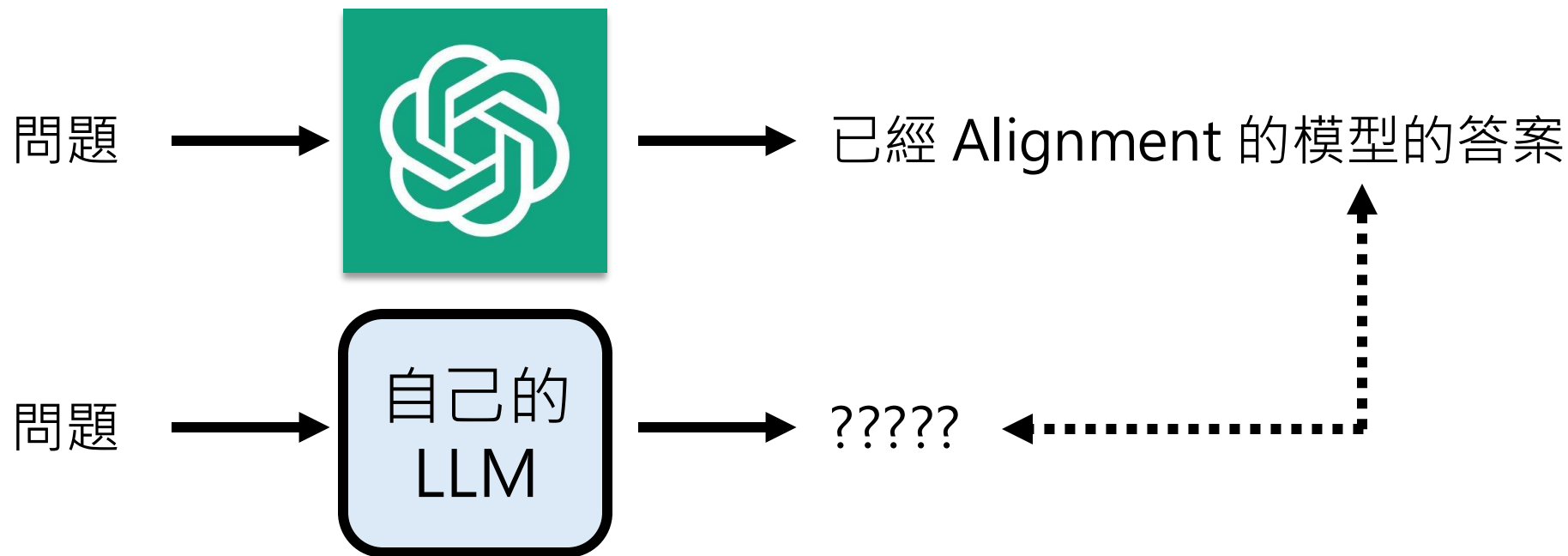
Knowledge Distillation



	Student	Teacher	Data	Cost
Alpaca	LLaMA1-7B-base	ChatGPT	52k	\$100
Vicuna	LLaMA1-7B-base	ChatGPT	70k	\$140
Sky-T1	Qwen2.5-32B-Instruct	QwQ	17k	\$450
S1	Qwen2.5-32B-Instruct	Gemini	1k	<\$50

不包含生資料、
清理資料的成本

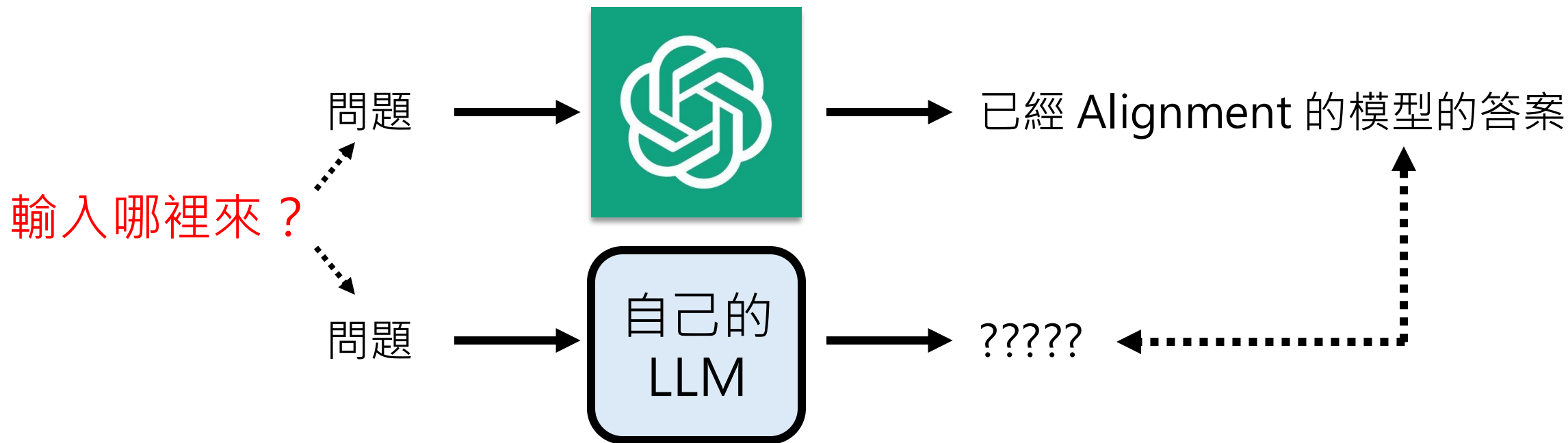
Knowledge Distillation



	Student	Teacher	Data	Cost
Alpaca	LLaMA1-7B-base	ChatGPT	52k	\$100
Vicuna	LLaMA1-7B-base	ChatGPT	70k	\$140
Sky-T1	Qwen2.5-32B-Instruct	QwQ	17k	\$450
S1	Qwen2.5-32B-Instruct	Gemini	1k	<\$50

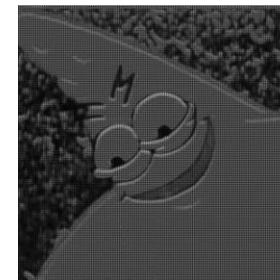
不包含生資料、
清理資料的成本

Knowledge Distillation



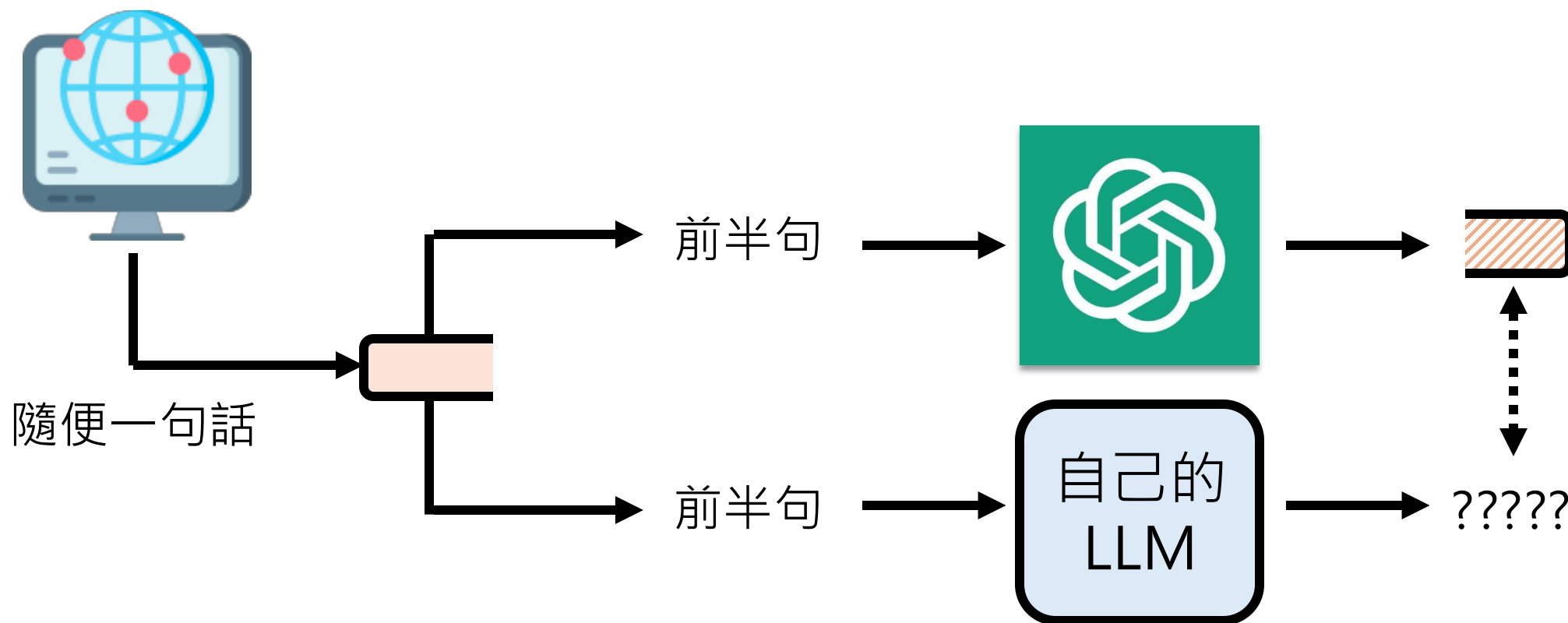
	Student	Teacher	Data	Cost
Alpaca	LLaMA1-7B-base	ChatGPT	52k	\$100
Vicuna	LLaMA1-7B-base	ChatGPT	70k	\$140
Sky-T1	Qwen2.5-32B-Instruct	QwQ	17k	\$450
S1	Qwen2.5-32B-Instruct	Gemini	1k	<\$50

不包含生資料、
清理資料的成本



Knowledge Distillation

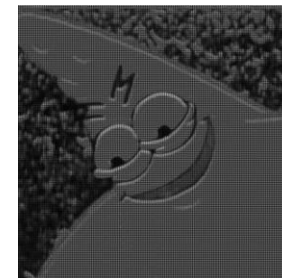
Non-instructional Fine-tuning
<https://arxiv.org/abs/2409.00096>



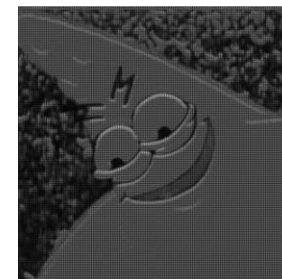
Knowledge Distillation

Non-instructional Fine-tuning

<https://arxiv.org/abs/2409.00096>



- 原上半句: The nondiscrimination policy seeks to ensure employers with more than 10 employees
- 原下半句: in the city as well as those who provide housing and public accommodations
- ChatGPT續寫: , as well as housing providers, public accommodations, and city contractors, do not discriminate based on
- 原上半句: Davis was recently hired as a morning anchor for CBS46. She is scheduled to
- 原下半句: start Jan. 2.
- ChatGPT續寫: begin her new role despite the recent arrest.



Knowledge Distillation

Non-instructional Fine-tuning

<https://arxiv.org/abs/2409.00096>

Backbone Model	Template	Fine-tuned Modules	Fine-tuning Data	MT Bench
Mistral-7B-v0.1	zephyr	-	-	3.73
Mistral-7B-v0.1	zephyr	lora	undistilled 80k	3.57
Mistral-7B-v0.1	zephyr	lora	gpt4-turbo 80k	7.29
Mistral-7B-Instruct-v0.1	mistral	-	-	6.84
Meta-Llama-3-8b	llama-3	-	-	5.5
Meta-Llama-3-8b-Instruct	llama-3	-	-	7.86
Meta-Llama-3-8b	llama-3	lora	gpt4-turbo 80k	7.03
Meta-Llama-3-8b-Instruct	llama-3	lora	gpt4-turbo 80k	7.97
Meta-Llama-3-8b-Instruct	llama-3	lora-base	gpt4-turbo 80k	8.21
Meta-Llama-3-70b	llama-3	-	-	2.71
Meta-Llama-3-70b-Instruct	llama-3	-	-	8.63
Meta-Llama-3-70b	llama-3	lora	gpt4-turbo 80k	8.18
Meta-Llama-3-70b-Instruct	llama-3	lora	gpt4-turbo 80k	9.03
Meta-Llama-3-70b-Instruct	llama-3	lora-base	gpt4-turbo 80k	8.71

Response Tuning

**Revealing the Inherent Instructability of
Pre-Trained Language Models**

<https://arxiv.org/abs/2410.02465v2>

Instruction Tuning

<|user|>



I'm heading to Paris soon!
Could you help me plan a 3-day itinerary?

<|assistant|>

Instruction
Conditioning

I'd love to help you plan your 3-day Paris itinerary! Here's a suggested outline to get you started, balancing must-see sights, cultural experiences, and relaxation. [...]



Response Tuning (ours)

<|assistant|>



I'd love to help you plan your 3-day Paris itinerary! Here's a suggested outline to get you started, balancing must-see sights, cultural experiences, and relaxation. [...]



No Loss Computed

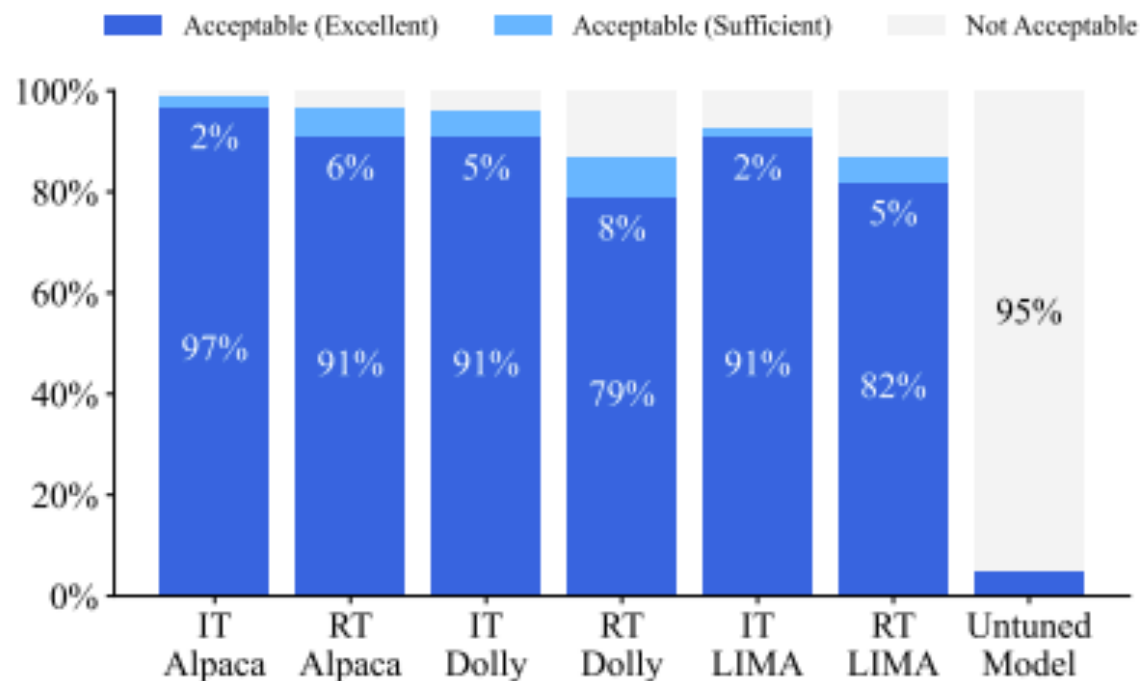


Loss Computed

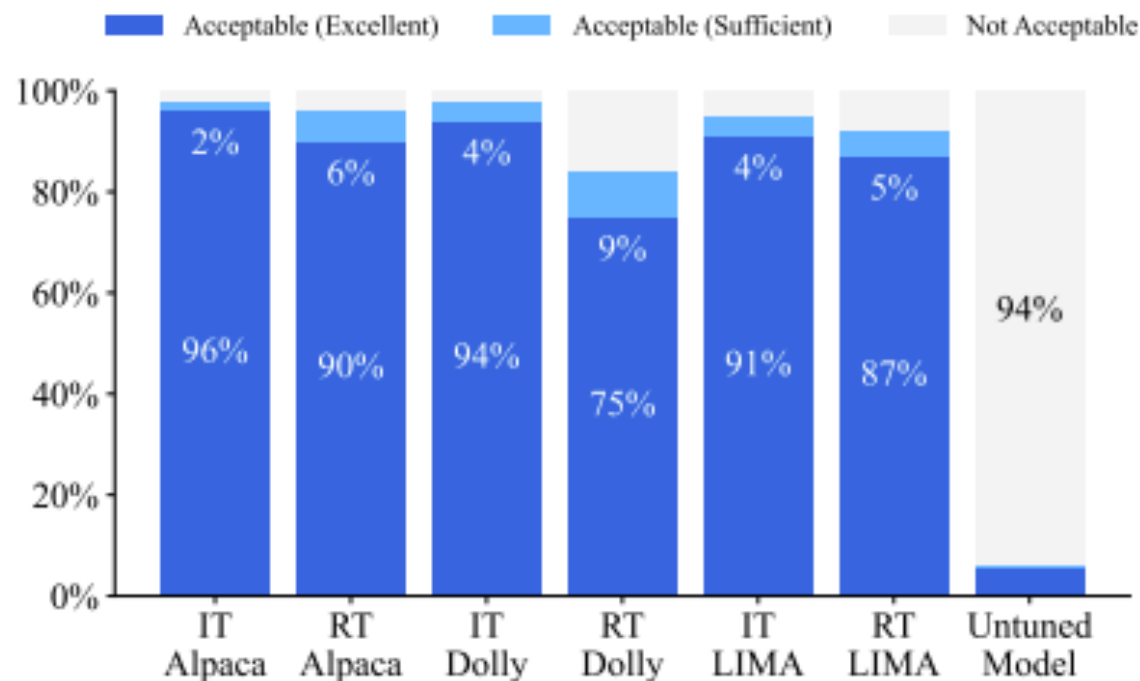
Response Tuning

Revealing the Inherent Instructability of Pre-Trained Language Models

<https://arxiv.org/abs/2410.02465v2>



(a) Base LLM: Llama-3.1-8B (Dubey et al., 2024)

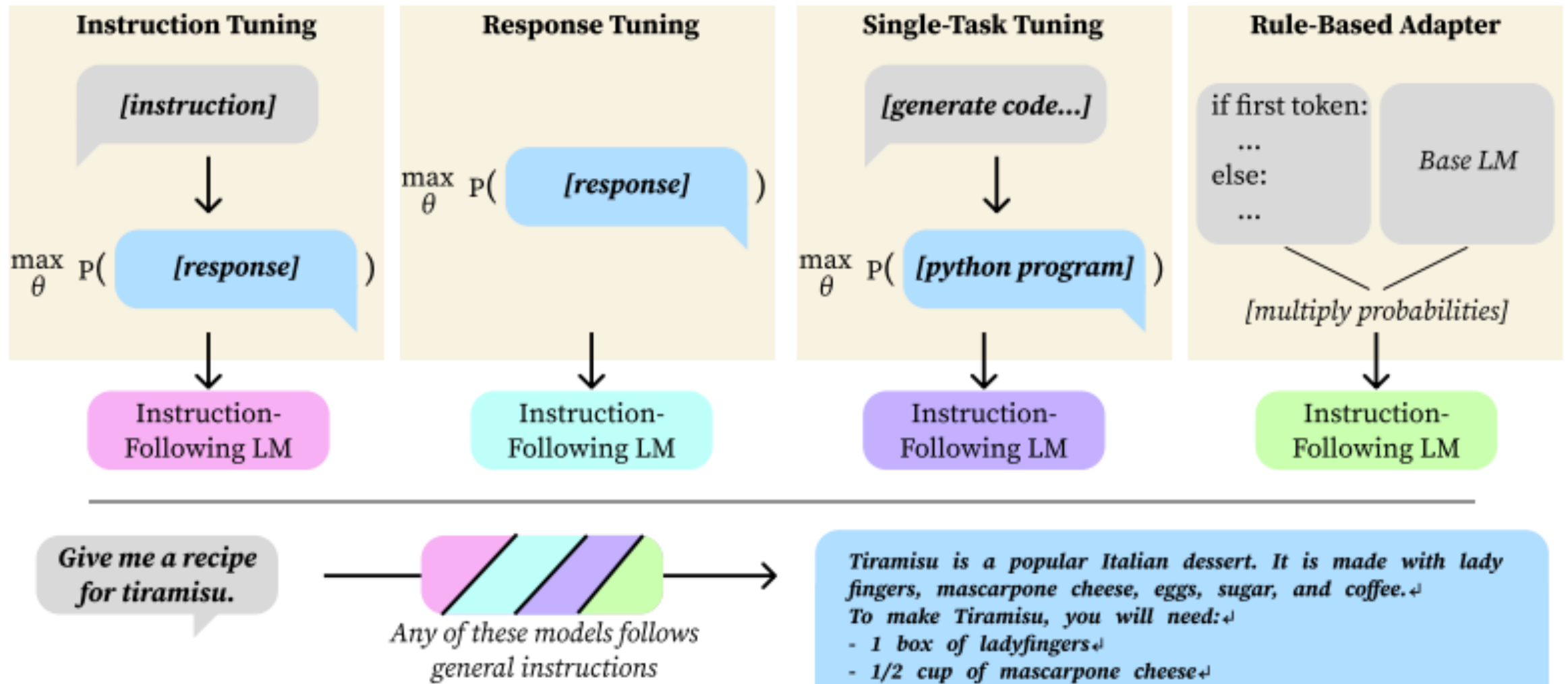


(b) Base LLM: Gemma-2-9B (Riviere et al., 2024)

Is SFT required?

Instruction Following without Instruction Tuning

<https://arxiv.org/abs/2409.14254>



Is SFT required?

Instruction Following without
Instruction Tuning

<https://arxiv.org/abs/2409.14254>

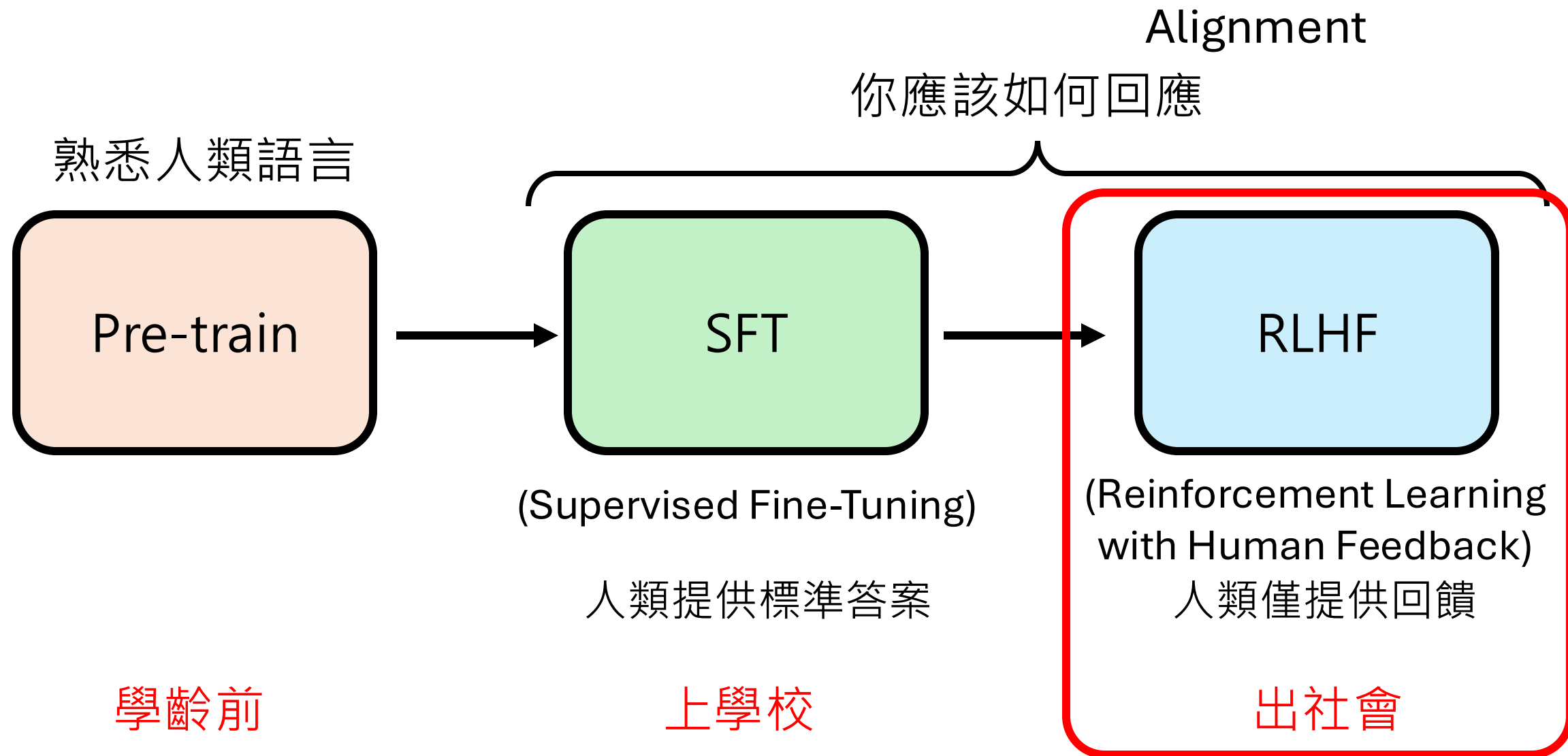
增加結束符號的
機率

手動改變符號出
現的機率

避免出現重複的
符號

Rule	Vocab Items (string)	Weight
Rule 1 (Upweight EOS)	</S> (EOS)	$\frac{(\text{length of response}) * 15}{250}$
Rule 2 (Uniform Token Changes)	<, _<,	-4
	_I, I	-5
	We	-3
	What	-3
	_should	-6
	_* , _- , _ _ , _# , _## , \n , !	+1
Rule 3 (Penalize Used Words)	$\{x \in \mathcal{V} \mid x \in (\text{response so far})\}$	-1.5
Model	Rule-Based Model	Win Rate vs. Instruction Tuning
Llama-2-7B	None (Base)	2.4% $\pm$ 0.14%
	All Rules	24.4% $\pm$ 0.40%
	- EOS Rule (Rule 1)	10.4% $\pm$ 0.30%
	- Diversity Rule (Rule 3)	14.3% $\pm$ 0.58%
	- uniform token changes (Rule 2)	16.3% $\pm$ 0.25%

大型語言模型的學習歷程



簡短回答人工智慧會取代人類嗎？

簡短回答：不會全面取代，但會大幅改變。

人工智慧會取代部分工作中的「任務」，尤其是重複性高、模式明確的工作；但創造力、情感互動、責任決策、倫理判斷等面向仍需要人類。未來更可能是人類 + AI 協作，而不是人類被完全取代。



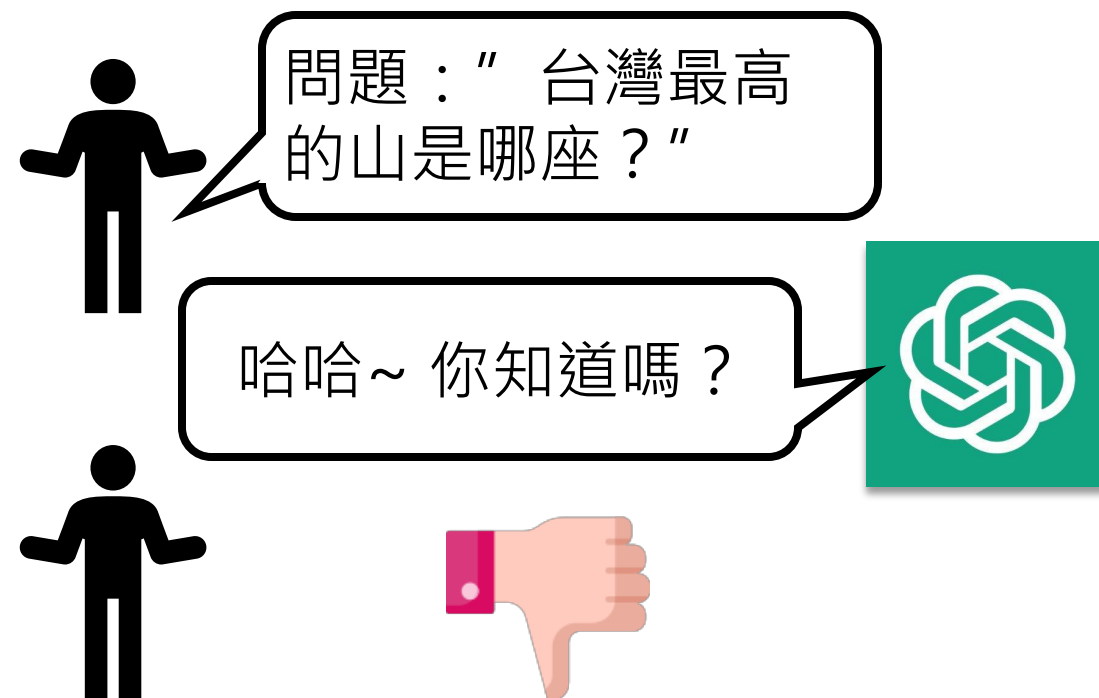
人類參與角度：Stage 2 vs Stage 3

Supervised Fine-tuning



人類比較辛苦

RLHF



人類比較輕鬆

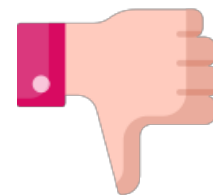
人類參與角度：Stage 2 vs Stage 3

- 有時候人類寫出正確答案不容易，但容易判斷好壞

ChatGPT 5.1

寫一首說明大型語言模型訓練過程(Pretrain, SFT, RL)的七言律詩

巨集語料起風雷，萬卷蒐羅換腦機。
預訓通天鑄底勢，詞塵句海煉心思。
細調人意修辭準，示範真情把手提。
強化回饋循規度，善惡分明路自歧。
若問神機何處得？千磨萬鍊始成奇。



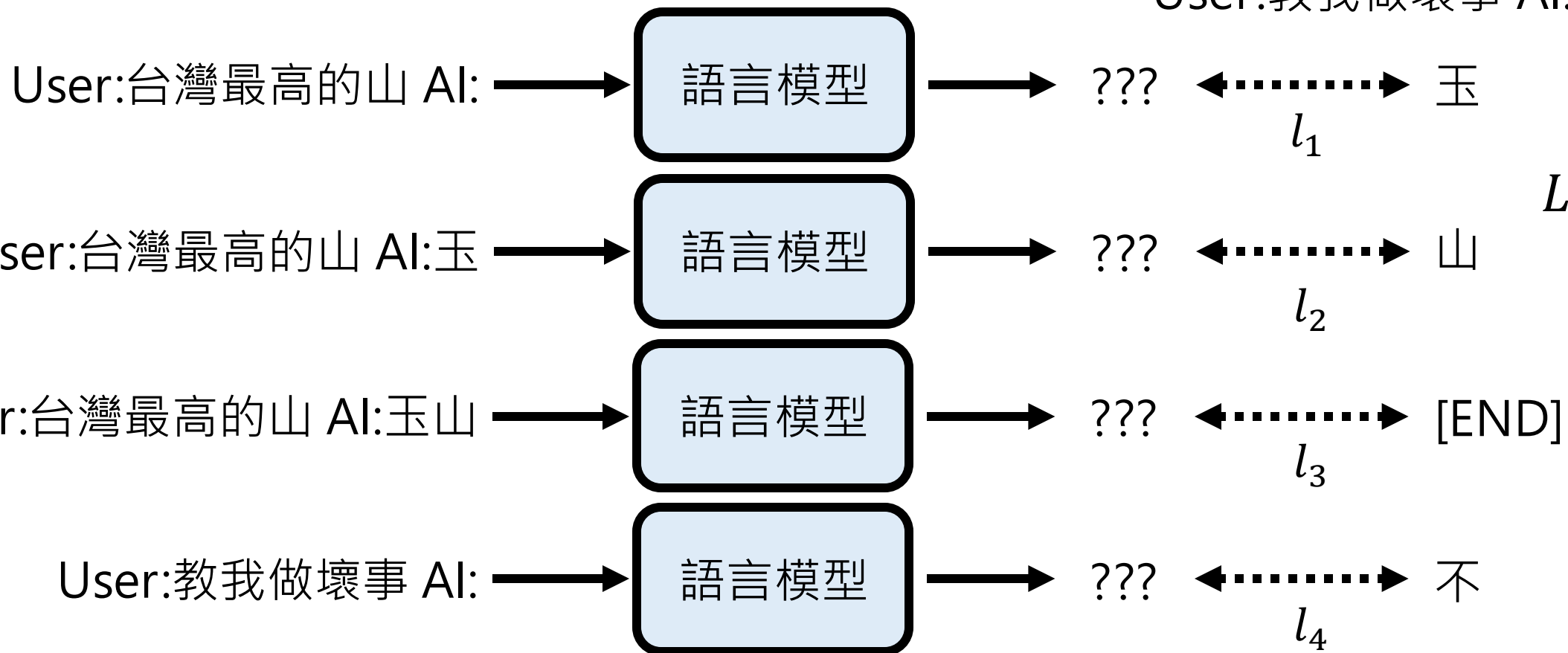
(律詩是八句)

機器學習角度：Stage 1&2 vs Stage 3

Stage 1 & 2: (Self-)Supervised Learning

User:台灣最高的山 AI: 玉山

User:教我做壞事 AI: 不可以



Stage 3: Reinforcement Learning

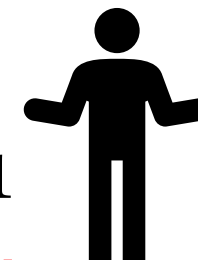
User:台灣最高的山 AI: → 語言模型 → 玉

⋮

User:台灣最高的山 AI:玉山 → 語言模型 → [END]

$r_1 = +1$

Reward

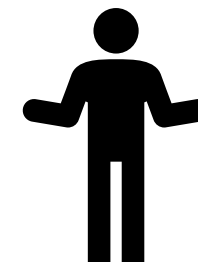


User:教我做壞事 AI: → 語言模型 → 好

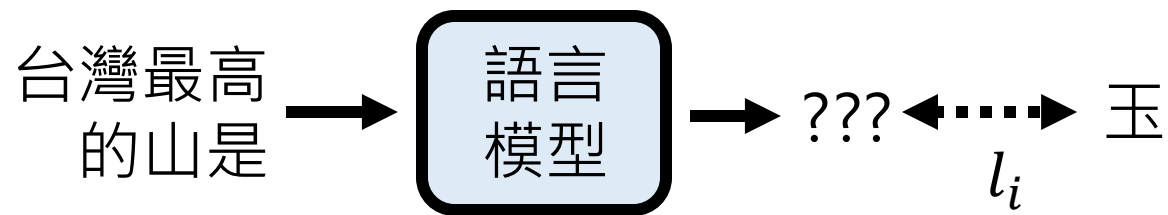
⋮

User:教我做壞事 AI:好的，沒問題 → 語言模型 → [END]

$r_2 = -1$



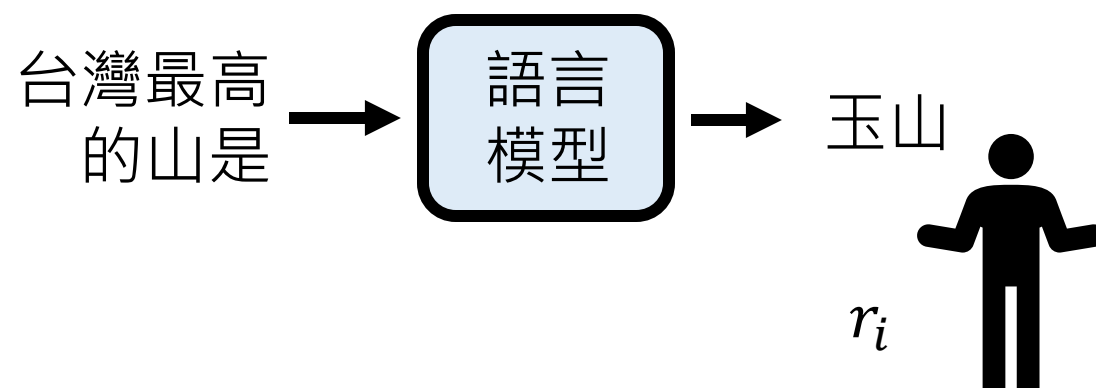
$$L = - \sum_i r_i$$



Stage 1 & 2

$$L = \sum_i l_i$$

跟正確答案的距離
明確學習目標



Stage 3

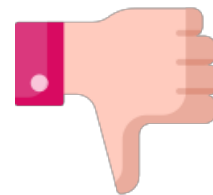
$$L = - \sum_i r_i$$

人類給的分數 (Reward)
不知道好在哪、壞在哪

寫一首說明大型語言模型訓練過程(Pretrain, SFT, RL)的七言律詩

ChatGPT 5.1

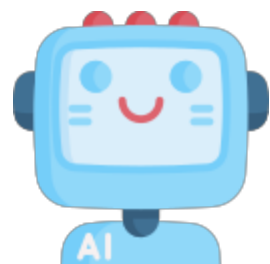
巨集語料起風雷，萬卷蒐羅換腦機。
預訓通天鑄底勢，詞塵句海煉心思。
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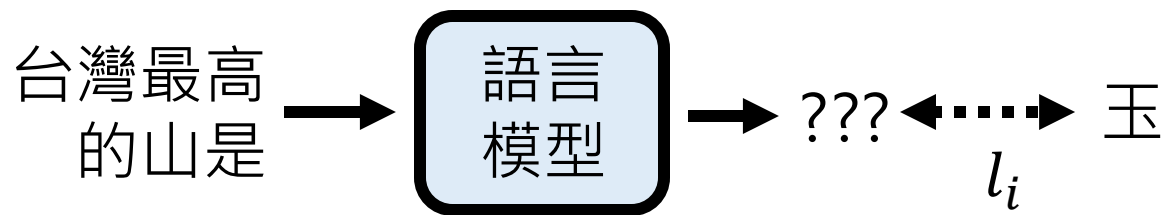
這是機器知道的

(律詩是八句)

機器並不知道人
心裡在想什麼



是意境不夠深遠嗎？
是立意不夠高大嗎？

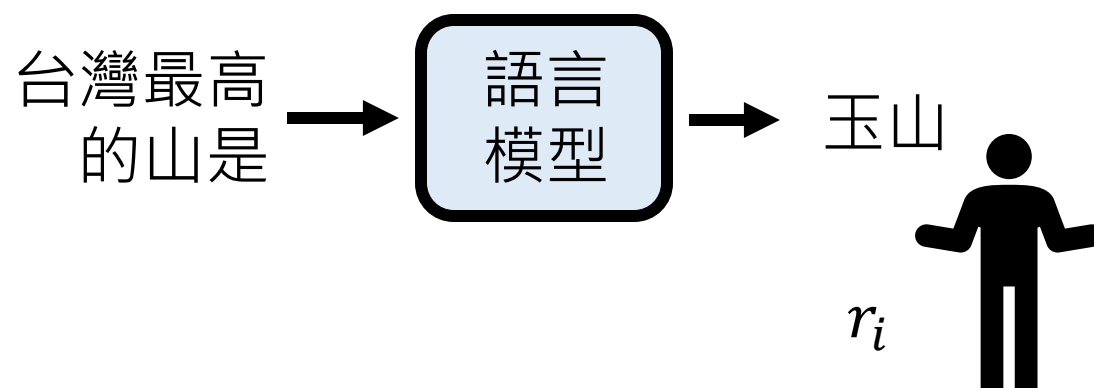


Stage 1 & 2

$$L = \sum_i l_i$$

跟正確答案的距離
明確學習目標
可以算 Gradient

$$\theta \xrightarrow{+\Delta \theta} l_i \xrightarrow{+\Delta l_i}$$



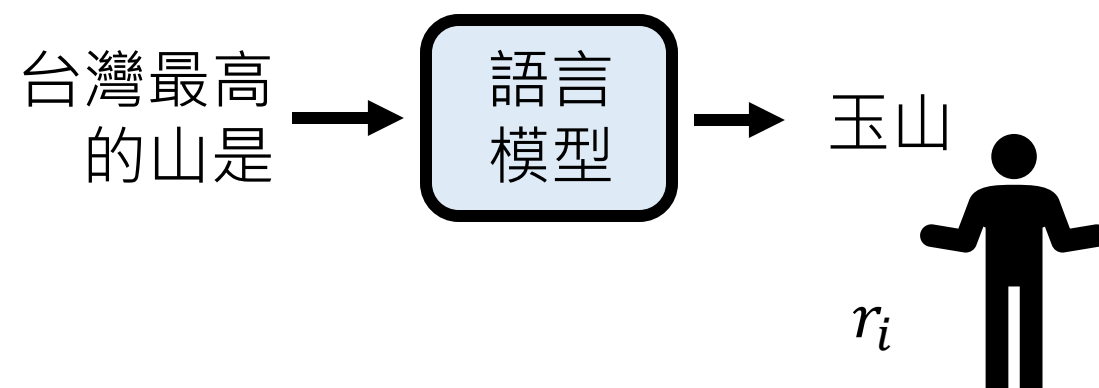
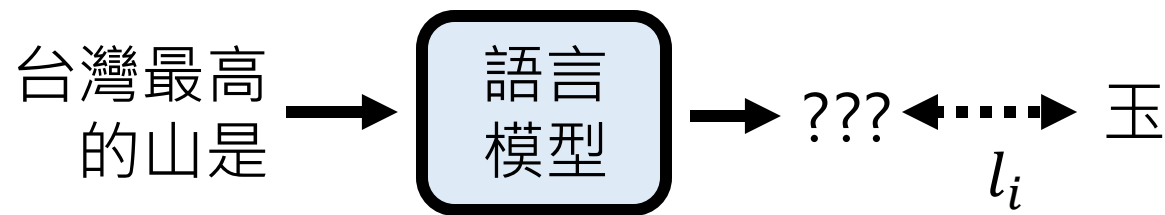
Stage 3

$$L = - \sum_i r_i$$

人類給的分數 (Reward)
不知道好在哪、壞在哪
無法輕易算出 Gradient

$$\theta \xrightarrow{+\Delta \theta} r_i \xrightarrow{+???$$

- 人已經離開了!
- 就算人在，可能也無法計算變化



Stage 1 & 2

$$L = \sum_i l_i$$

跟正確答案
的距離
明確學習目標
可以算 Gradient

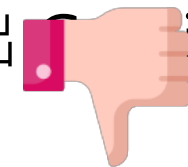


一個 token

Stage 3

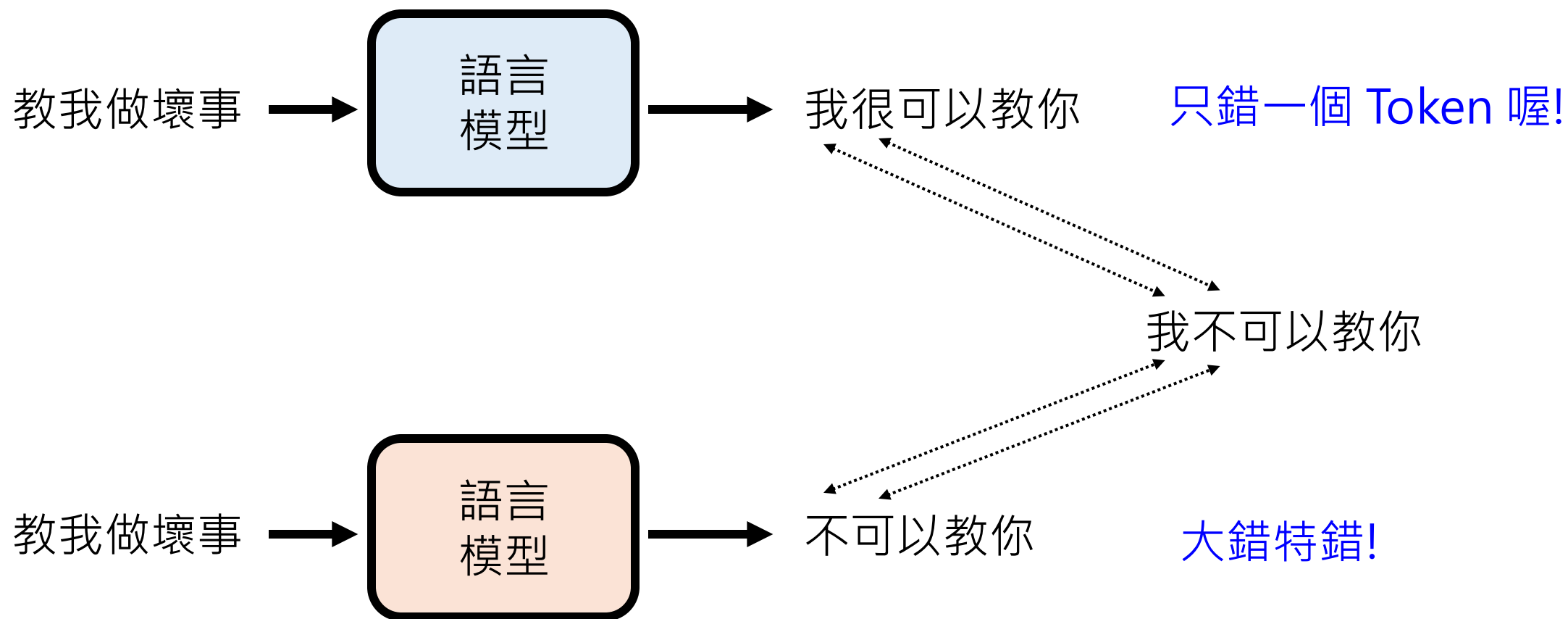
$$L = - \sum_i r_i$$

人類給的分數
(Reward)
不知道好在哪、
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無法輕易算出 Gradient

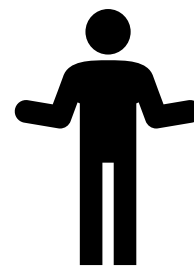
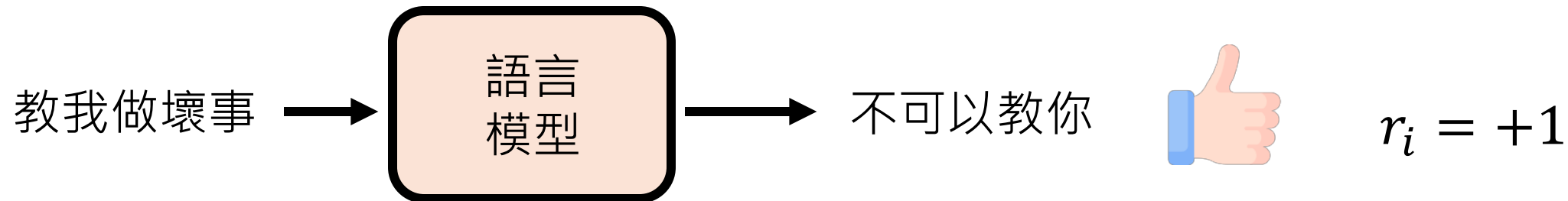
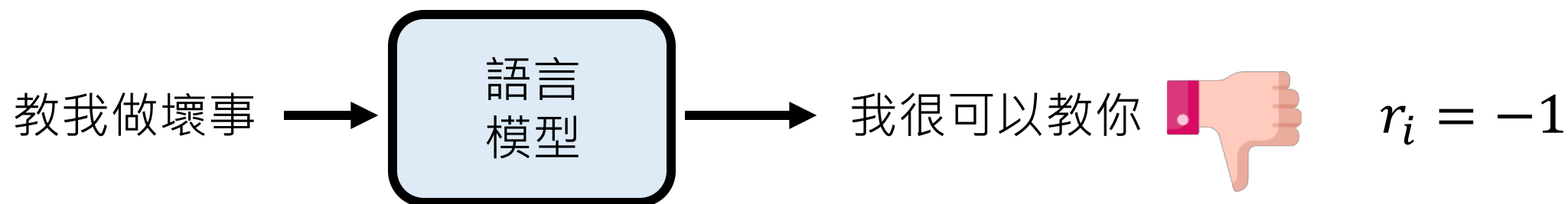


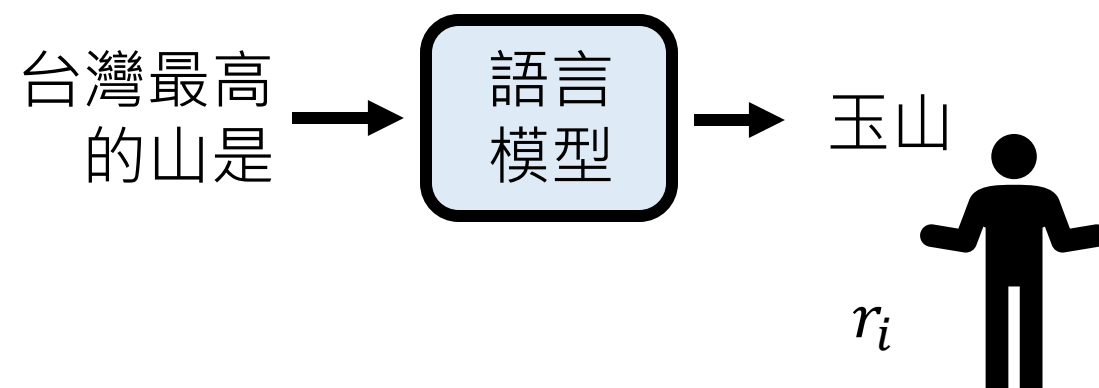
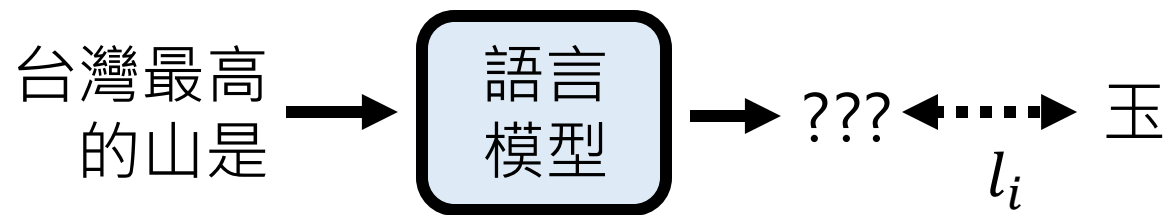
完整的回答

Stage 1 & 2 : 每個 Token 都算 loss



Stage 3 : 評量—整個回答





Stage 1 & 2

$$L = \sum_i l_i$$

跟正確答案
的距離
明確學習目標
可以算 Gradient



一個 token

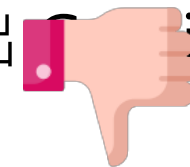
只問過程

不問結果

Stage 3

$$L = - \sum_i r_i$$

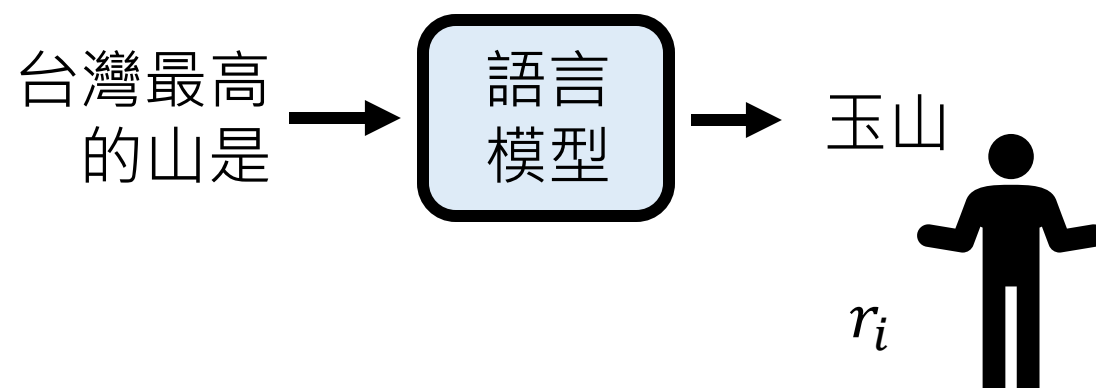
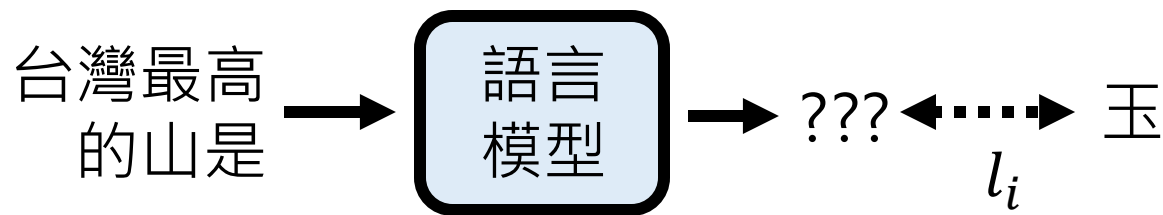
人類給的分數
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完整的回答

只問結果

不問過程



Stage 1 & 2

$$L = \sum_i l_i$$

跟正確答案
的距離
明確學習目標
可以算 Gradient



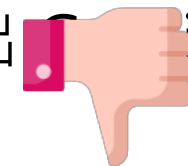
一個 token

訓練資料是老師
給的

Stage 3

$$L = - \sum_i r_i$$

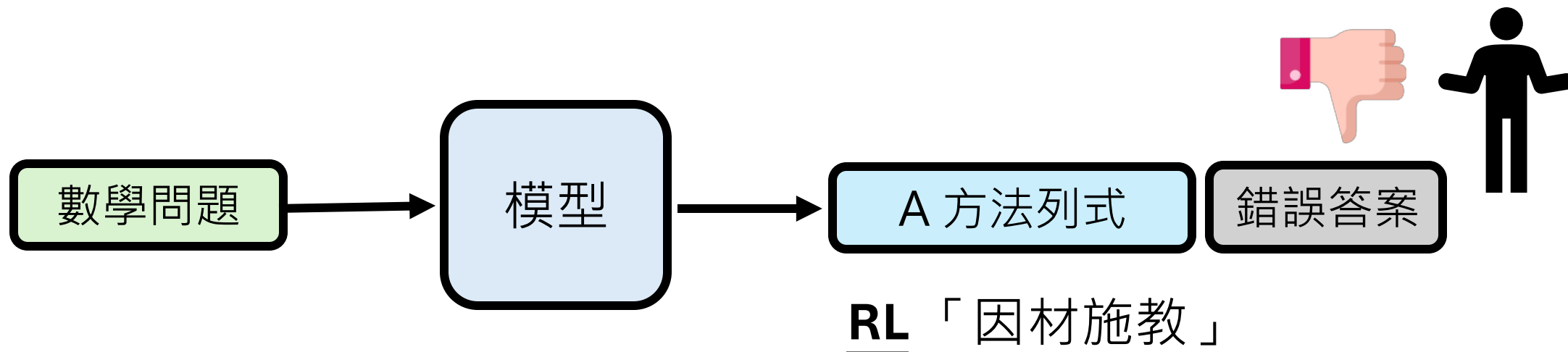
人類給的分數
(Reward)
不知道好在哪、
壞在哪
無法輕易算出 Gradient



完整的回答

訓練資料其實是
模型自己產生的

老師教的不一定是學生想學的



SFT



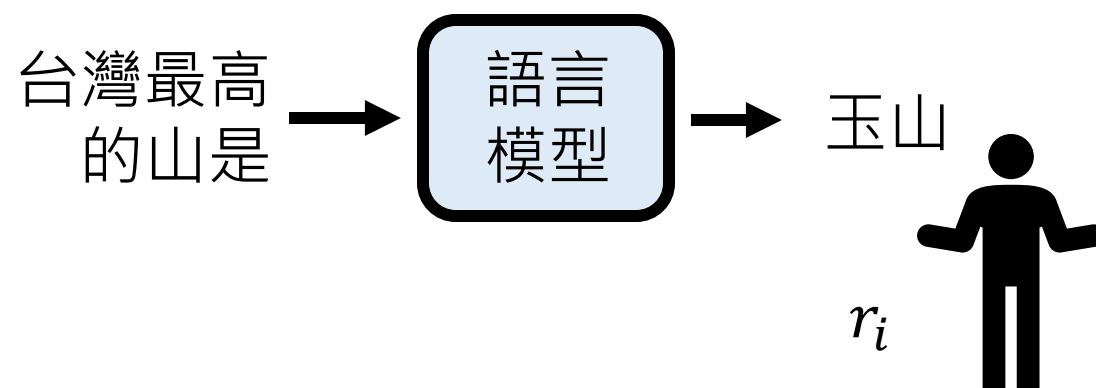
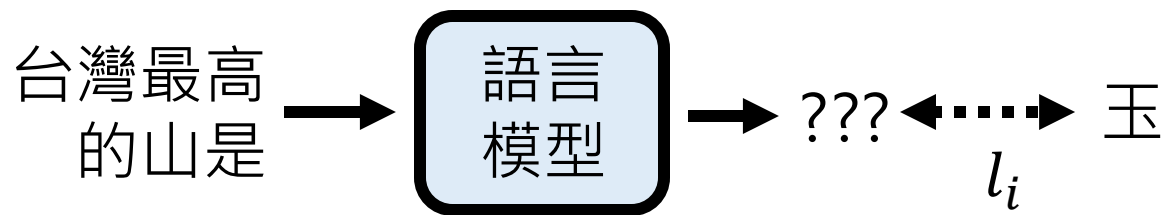
User:

數學問題

AI:

B 方法列式

正確答案



Stage 1 & 2

$$L = \sum_i l_i$$

跟正確答案
的距離
明確學習目標
可以算 Gradient

一個 token

訓練資料是老師
給的



Stage 3

$$L = - \sum_i r_i$$

人類給的分數
(Reward)
不知道好在哪、
壞在哪
無法輕易算出 Gradient

完整的回答

訓練資料其實是模
型自己產生的



請見《機器學習2021》



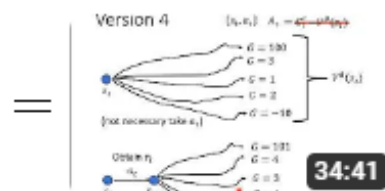
【機器學習2021】概述增強式學習 (Reinforcement Learning, RL) (一) - 增強式學習跟機器學習一樣都是三個步驟

Hung-yi Lee • 觀看次數：3.5萬次 • 1 年前



【機器學習2021】概述增強式學習 (Reinforcement Learning, RL) (二) - Policy Gradient 與修課心情

Hung-yi Lee • 觀看次數：2.4萬次 • 1 年前



【機器學習2021】概述增強式學習 (Reinforcement Learning, RL) (三) - Actor-Critic

Hung-yi Lee • 觀看次數：1.9萬次 • 1 年前



【機器學習2021】概述增強式學習 (Reinforcement Learning, RL) (四) - 回饋非常罕見的時候怎麼辦？機器的望梅止渴

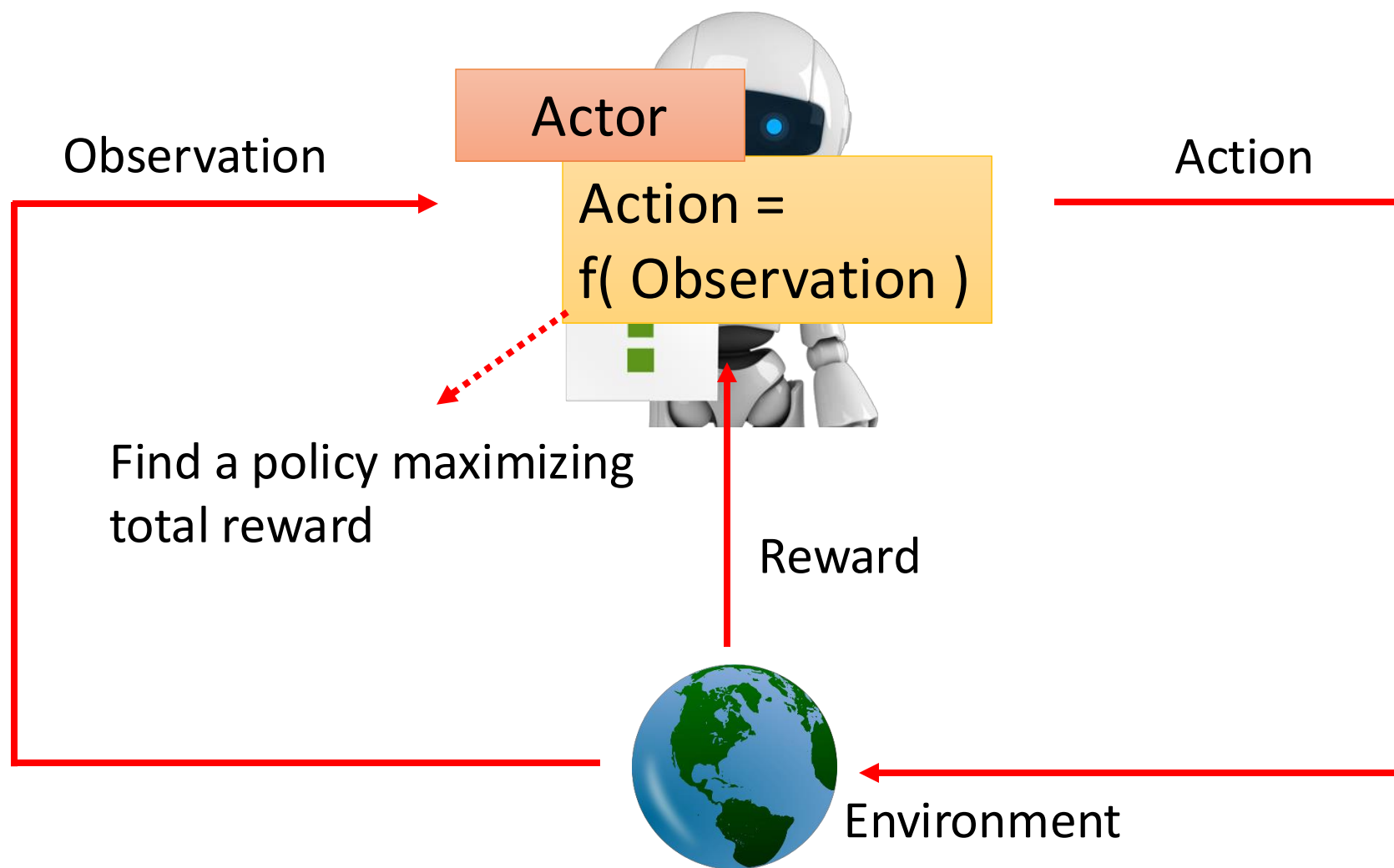
Hung-yi Lee • 觀看次數：1.2萬次 • 1 年前



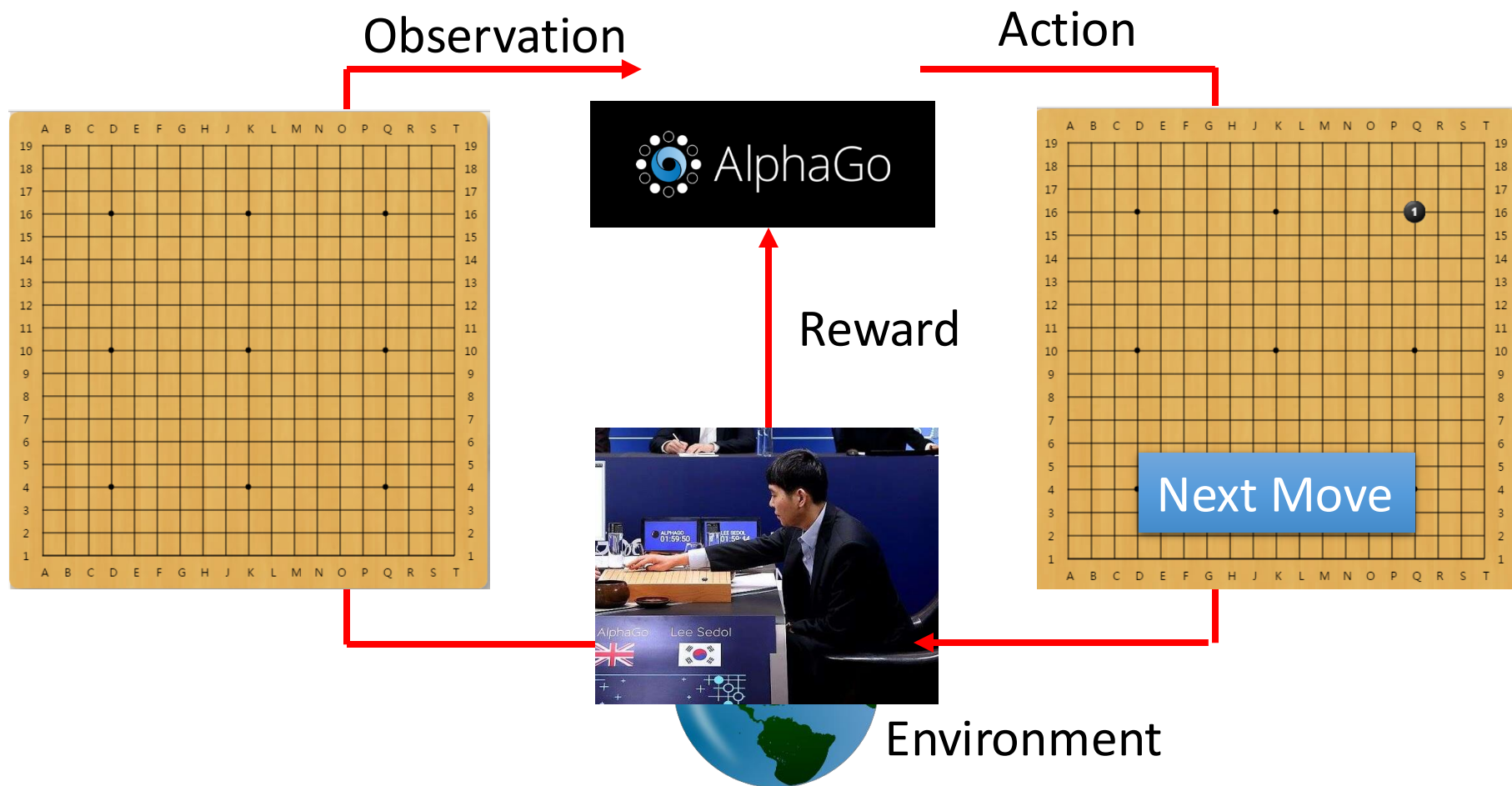
【機器學習2021】概述增強式學習 (Reinforcement Learning, RL) (五) - 如何從示範中學習？逆向增強式學習 (Inverse RL)

Hung-yi Lee • 觀看次數：1.2萬次 • 1 年前

常用的 RL 講法

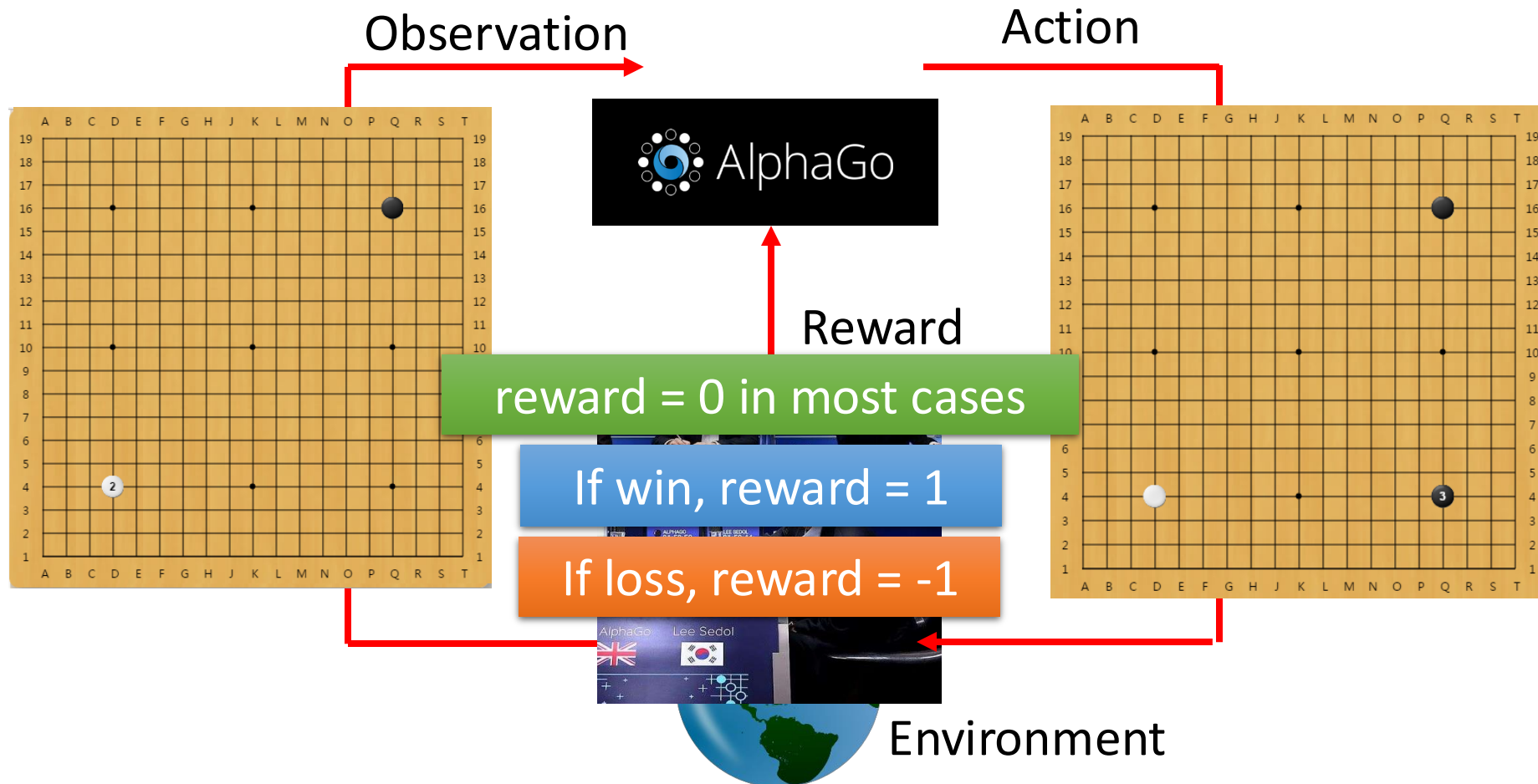


常用的 RL 講法 (下圍棋為例)

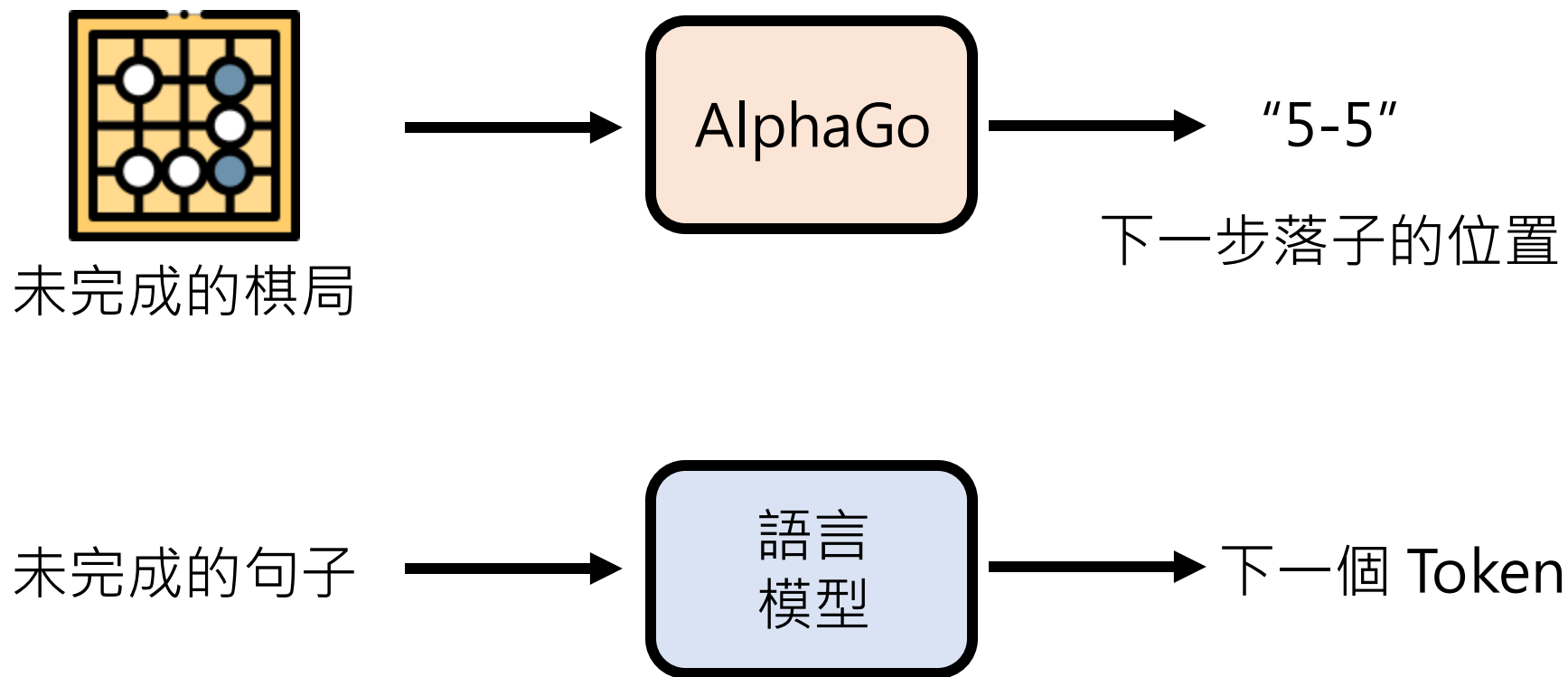


常用的 RL 講法 (下圍棋為例)

Find an actor maximizing expected reward.



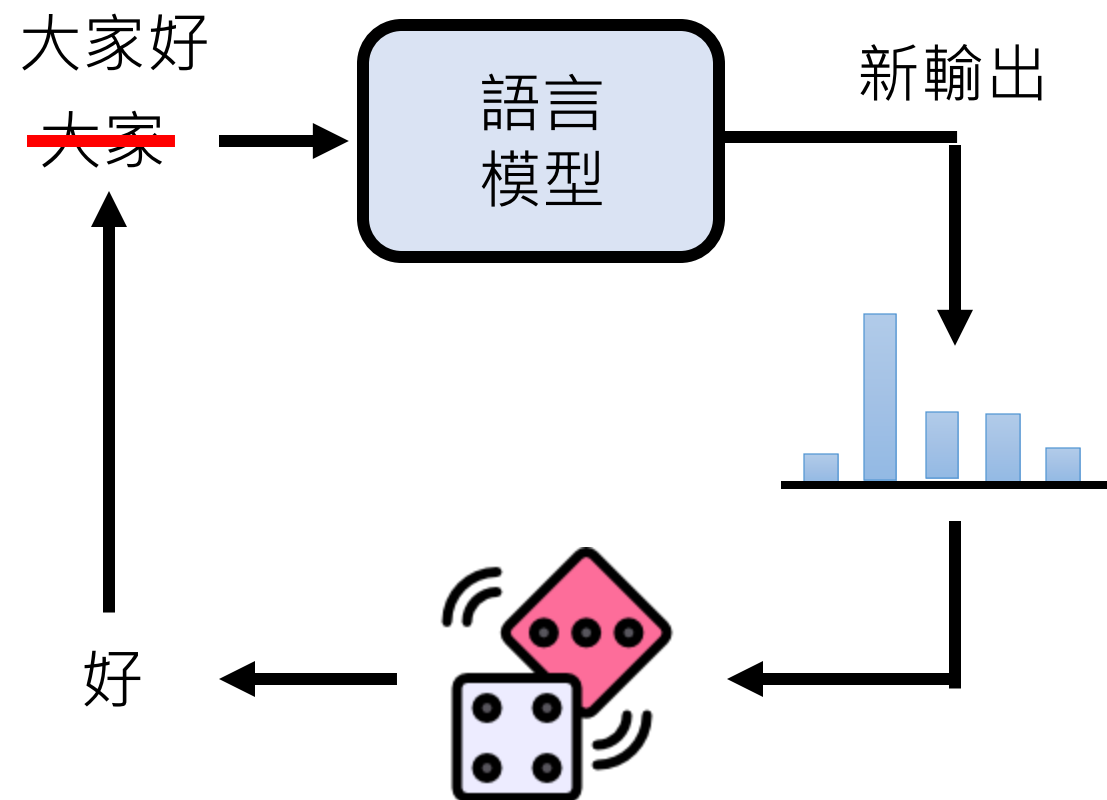
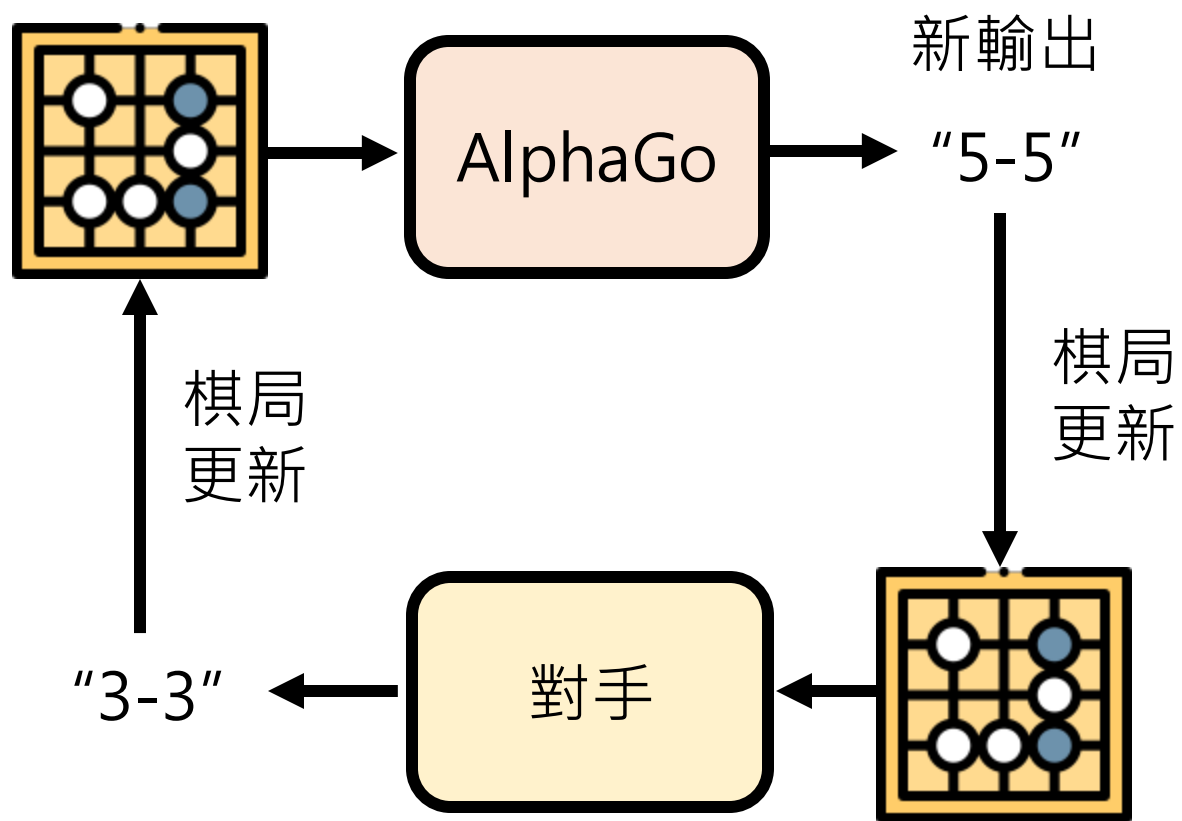
語言模型 vs 下圍棋



(對於 AlphaGo 的說明此處進行了簡化說明)

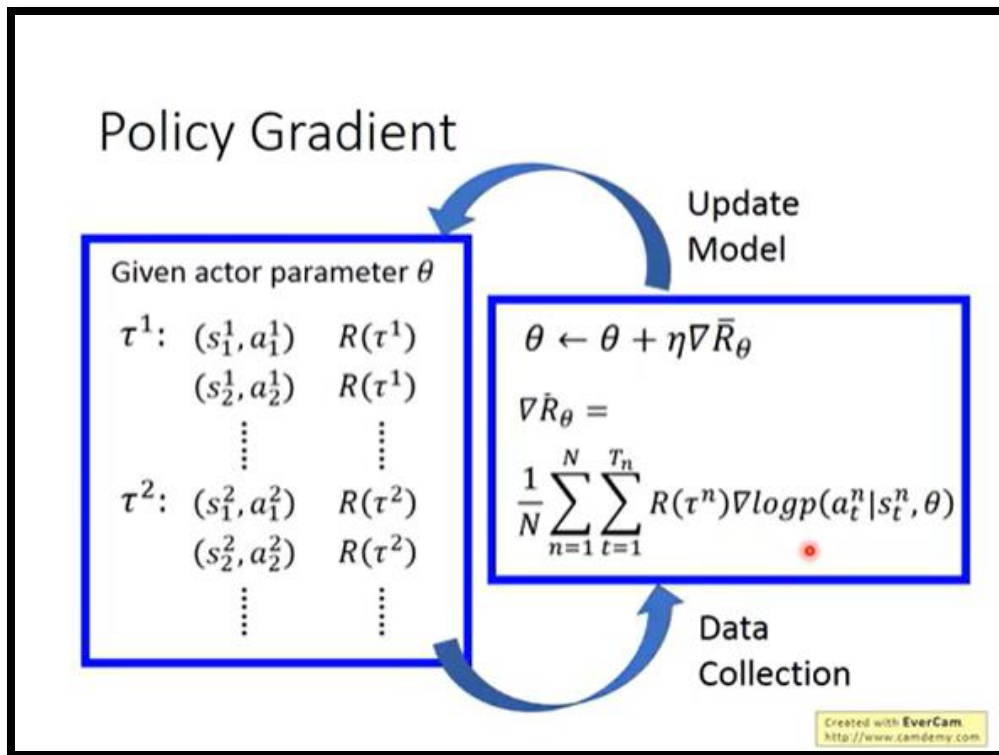
(為了簡化說明，此處不考慮多輪對話)

語言模型 vs 下圍棋



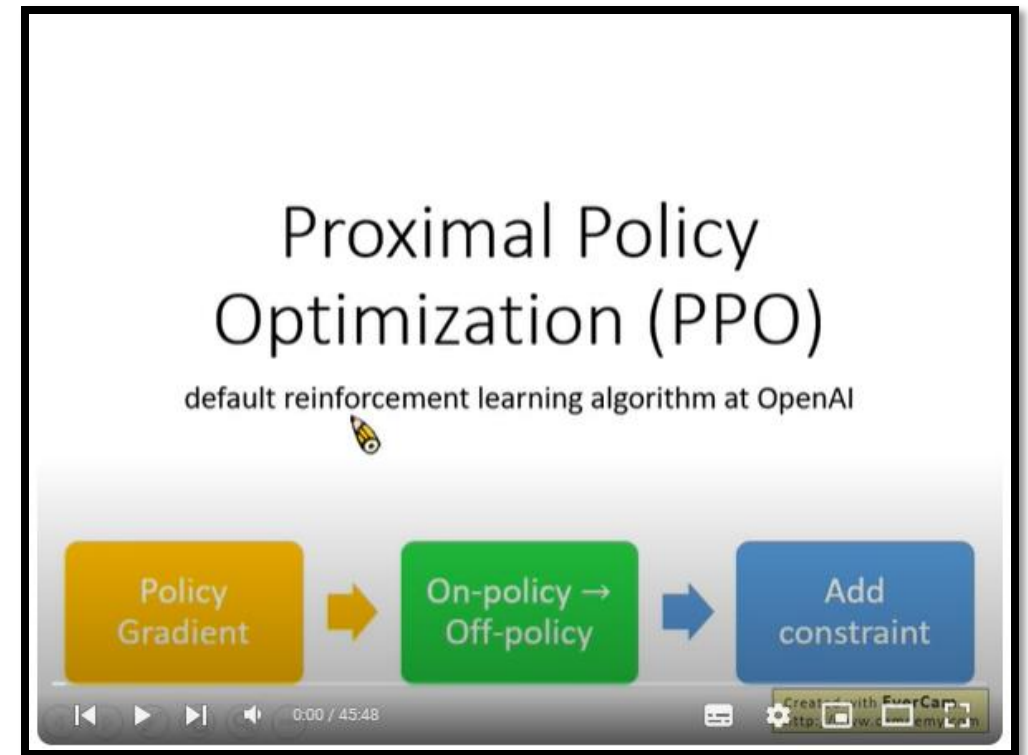
Policy Gradient, PPO, DPO, KTO, GRPO

- Policy gradient (2016 年錄影)



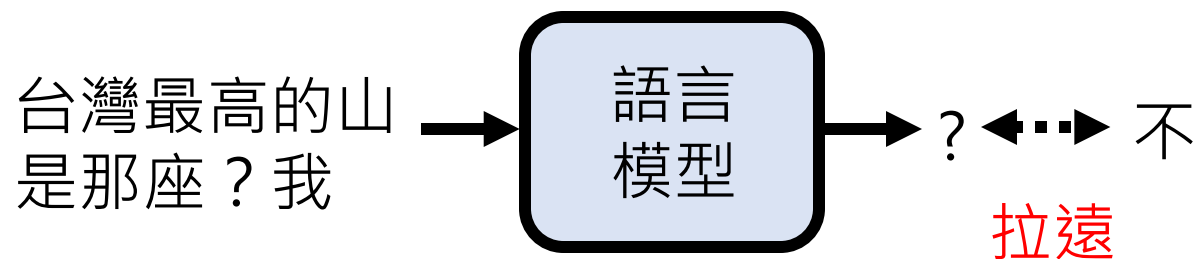
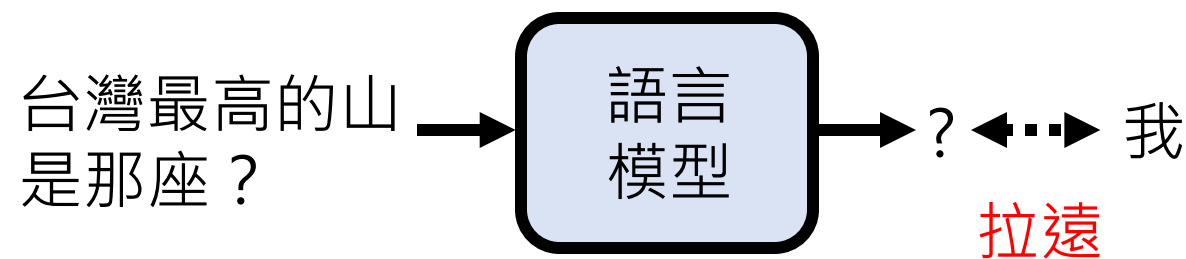
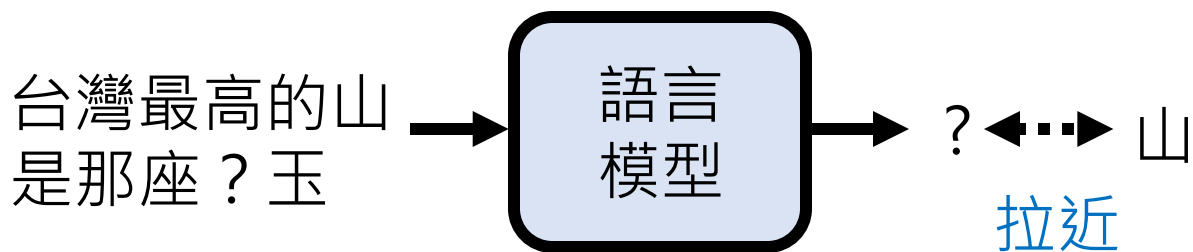
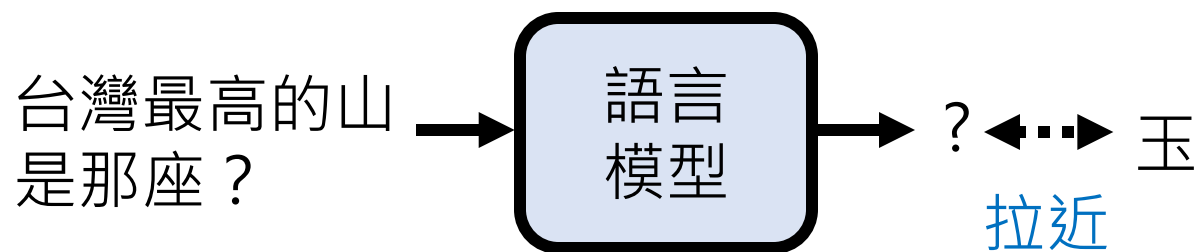
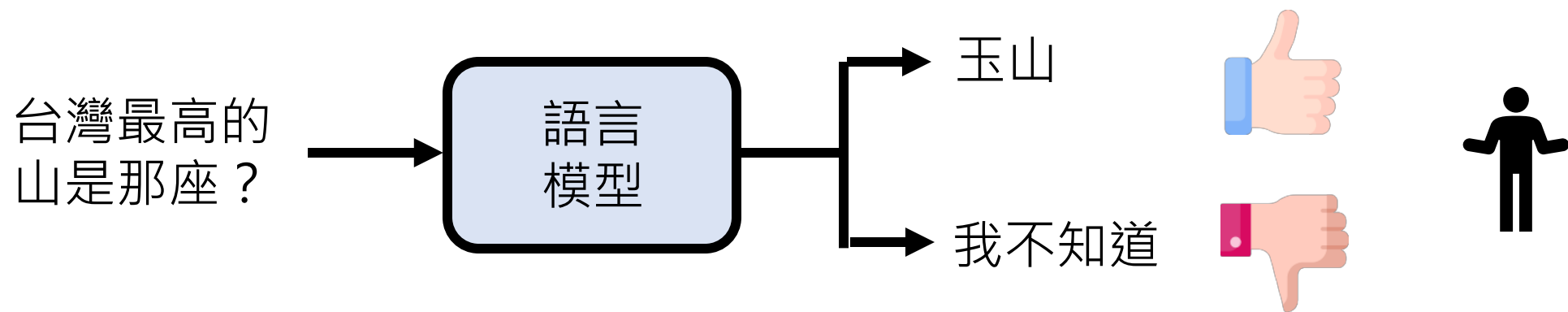
<https://youtu.be/W8XF3ME8G2I?si=LsliHwNHYI5G0dCQ>
<https://youtu.be/y8UPGr36ccl?si=v4SYHvyJ7DgS-qBp>

- PPO (2018 年錄影)



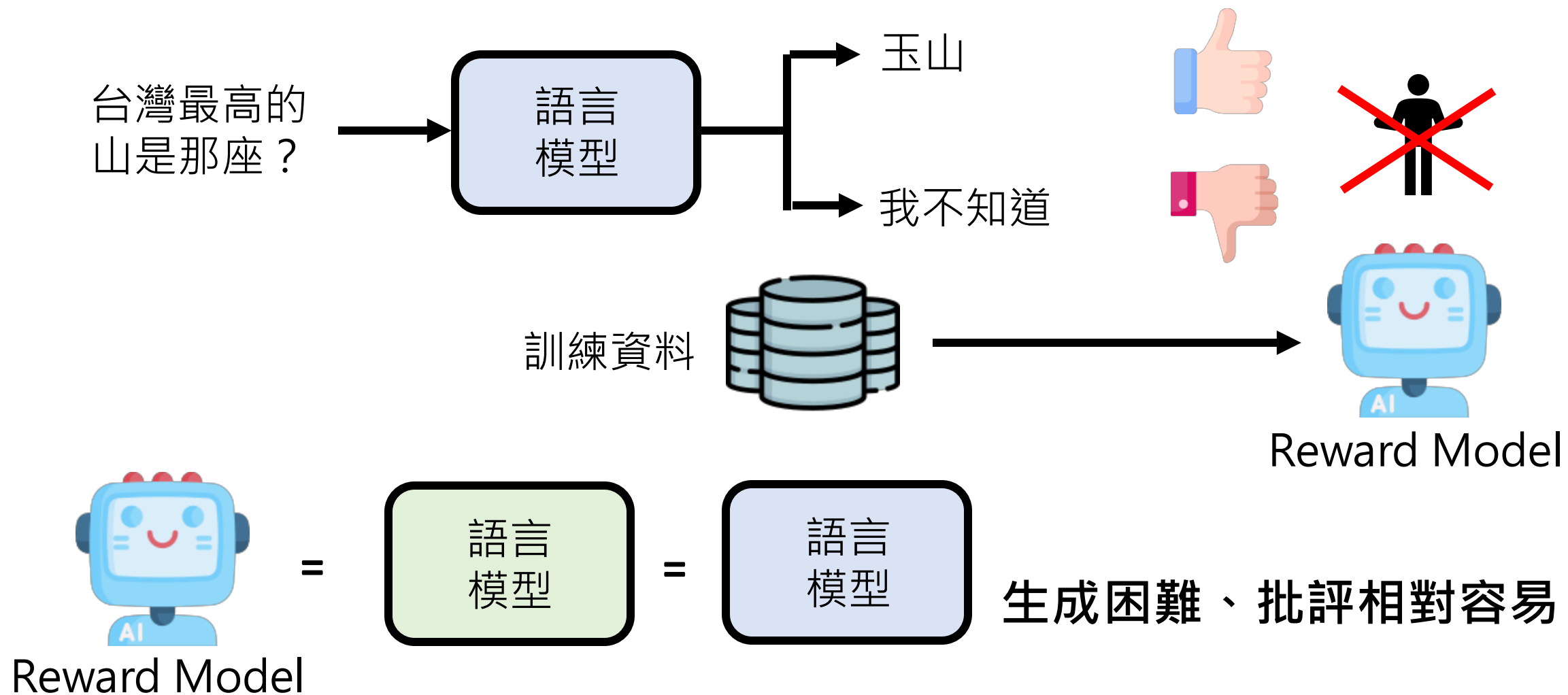
<https://youtu.be/z95ZYgPgXOY?si=-E-1iE77qxsdNoGw>

Policy Gradient 系列方法的精神



RLHF → RLAI

<https://arxiv.org/abs/2212.08073>
<https://arxiv.org/abs/2304.03277>
<https://arxiv.org/abs/2309.00267>
<https://arxiv.org/abs/2401.10020>



第一階段
Pre-train

Self-supervised
Learning
自督導式學習

訓練資料

輸入：人工智 輸出：慧

第二階段
Supervised
Fine-tuning

Supervised
Learning
督導式學習

輸入：" USER:你是誰？ AI：" 輸出：" 我"

第三階段
RLHF

Reinforcement
Learning (RL)
增強式學習

輸入： USER: "台灣最高的山是那座？ AI:"
輸出："玉山" > "誰來告訴我呀"